Biology

Unit 1:
molecules, transport and health

circulation:

Why do (Large organisms) need circulatory system?

i) Large animals have small surface area to volume ratio-

ii) Large animals have high metabolic rate

iii) so diffusion is not enough to transport oxygen and nutrients to cells.

iv) circulation is needed to transport oxygen and nutrients and to remove waste products like CO_2 and urine

-Why do (small organisms) not need a transport system?

i) they have small surface area to volume ratio.

ii) they have low metabolic rate.

iii) so, diffusion is efficient enough to transport oxygen} nutrients and remove
waste products.

- what are the properties of a good transport system?

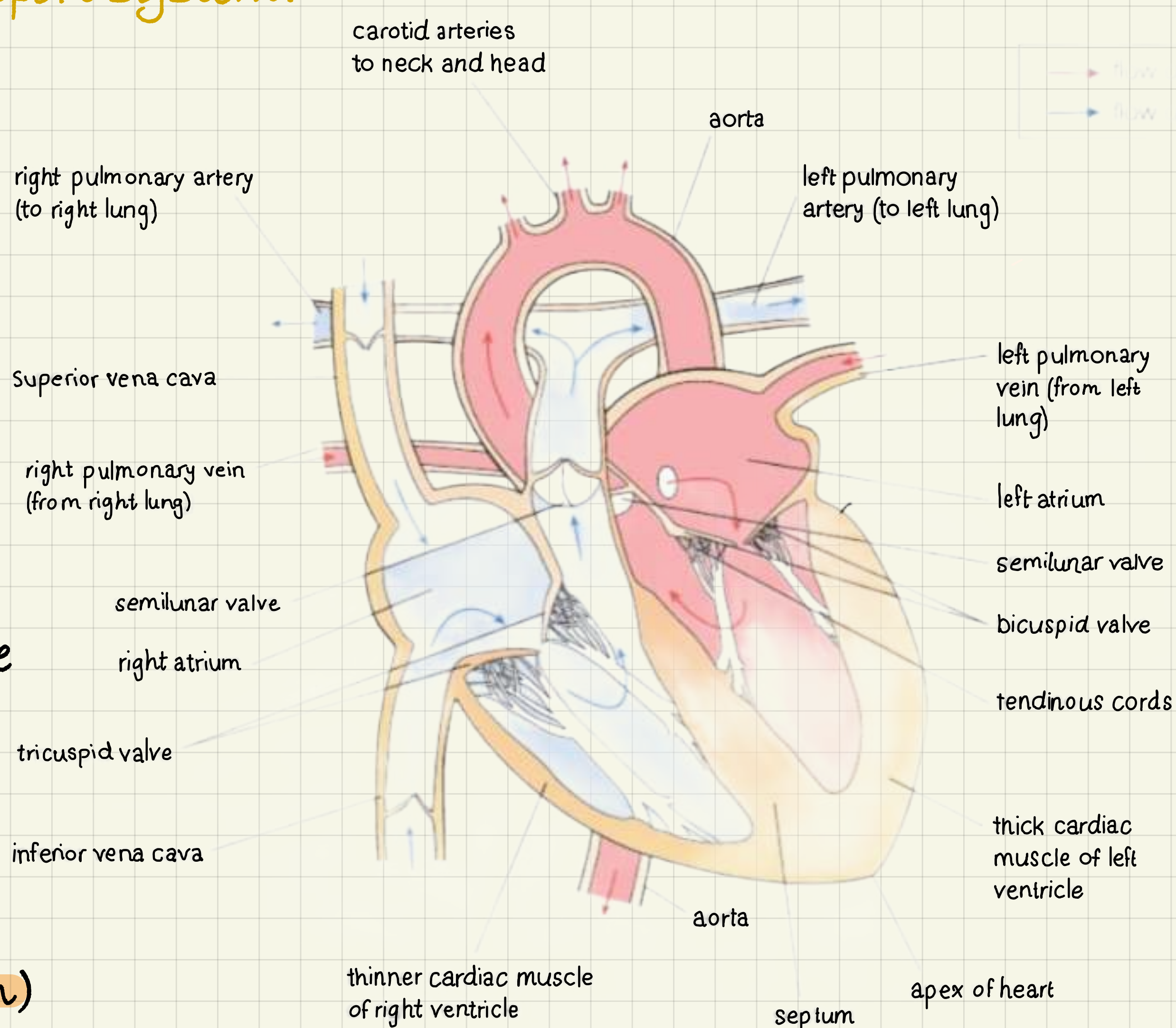
1 A pump → Heart

2 A pathway → Blood vessels

3 A medium → Blood

1 The heart:

- Is made from cardiac muscle.
- Made up from 4 champers
  - ↳ 2 upper:atria
  - ↳ 2 lower: ventricles
- Left side is separated from right side by a thick muscle named "Septum"
- 2 set of Atrioventricular values (between Atria and ventricles)
- 2 set of semilunar values (At base of aorta and pulmonary vein)
- ventricular walls are thinner than atrial walls
- the left ventricle is thicker than right ventricle.
- pulmonary vein (from lung)/artery (to lung).
- Main and largest vein: vena cava (to heart)
- Main artery: aorta (away from heart)
- pacemaker (nerve tissue) to regulate heart-beat



* stages of cardio cycle:

→ atrial systole :

1) deoxygenated blood reaches the heart, from vena cava from the body, and oxygenated blood from pulmonary vein from lungs, to atria.
2) Atria contracts and ventricle relaxes, creating high pressure at Atria; opening atrioventricular valves; pushing blood from atria to ventricle. (semilunar values are closed)

- in summary: atria contracts, ventricle relax, A.V value open, semilunar value close (Blood enters heart)

→ Ventricular systole:

1) atria relax, ventricle filled with blood contracts, creating high blood pressure in ventricles; pushing deoxygenated blood through pulmonary vein (to lungs) and oxygenated blood (to body)
through aorta. Semilunar valves are open and A.V valves are closed.

- in summary: Ventricle contract, atria relax, A.U values closed, semilunar open (blood pushed out of heart)

→ Diastole:

-Atria relax, ventricle relax, A.V value open, semilunar valve closed (to receive blood)

* importance of values:

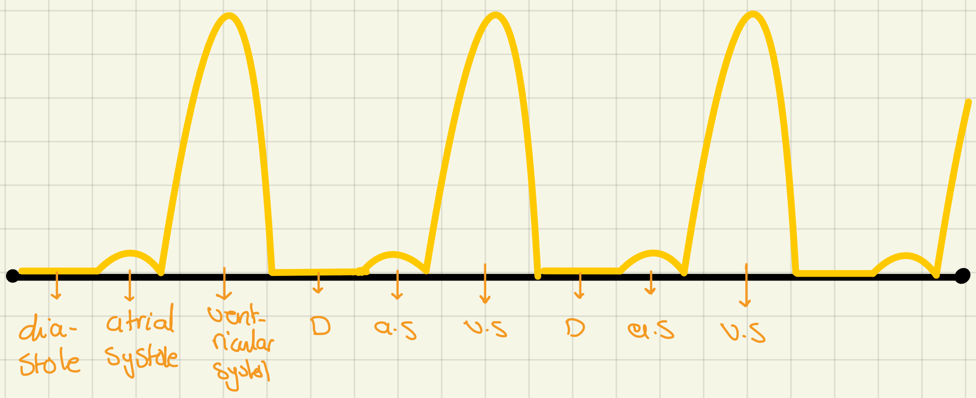
Atrioventricular values: to prevent the back-flow of blood from ventricle to atria during ventricular systole; as this would create a weaker circulation.

—values have a tough tendinous cord (heart string) to make sure values are not turned outside out by pressure exerted from ventricular contractions

* one full cardio cycle need 0.8 sec on average

→ calculate average heart rate?

60/0.8 = 75 bpm



* importance of double circulation:

① prevent mixing of oxygenated and deoxygenated blood; so more efficient supply of oxygen is delivered to body cells.

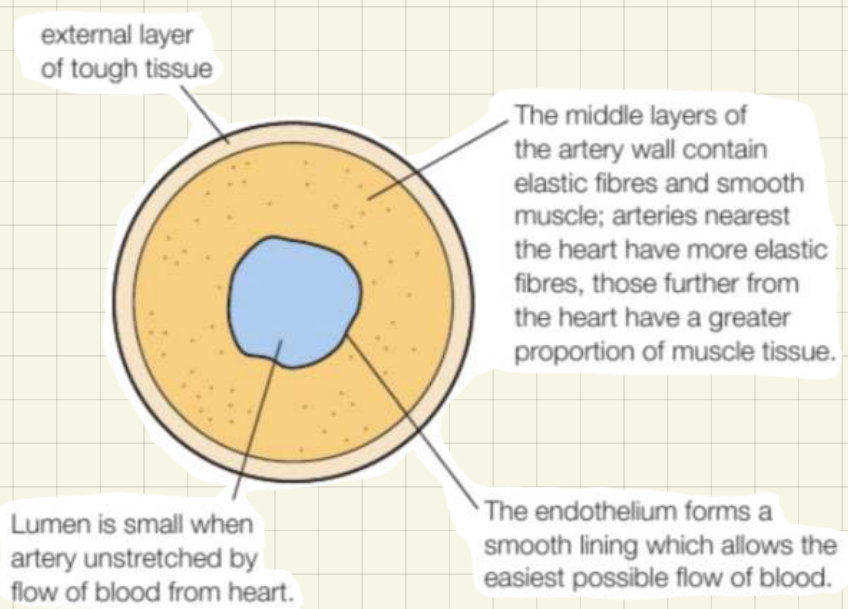
② to pump blood under high pressure to body cells's from left ventricle as it travels a longer distance than when it travels to the lungs from right ventricle's, as high blood pressure might damage delicate lung tissue

(pump blood at different pressure)

2 Blood vessels:

A) arteries:

↳ made up from 3 layers:
  ↳ smooth endothelium
  ↳ lumen
  ↳ externa, middle, inner tonica.



- they have strong/thick muscular walls to withstand high blood pressure
- have elastic wall to stretch and recoil.
- have coclegeng a type of potion, to give strength to withstand high blood pressure
- smooth endothelium lining to maintain high blood pressure

* carry oxygenated blood away from heart except for:

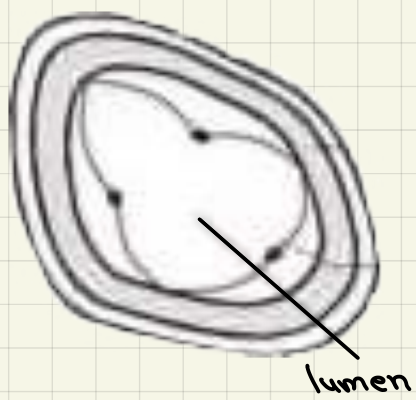
pulmonary artery → deoxygenated blood from heart to lung

umbilical artery → deoxygenated blood from fetus to placenta

B) veins:

↳ made up from the same artery layers but with wider lumen

- have blood value to prevent back flow of blood
- have thinner muscular walls than artery walls (no high pressure to maintain)



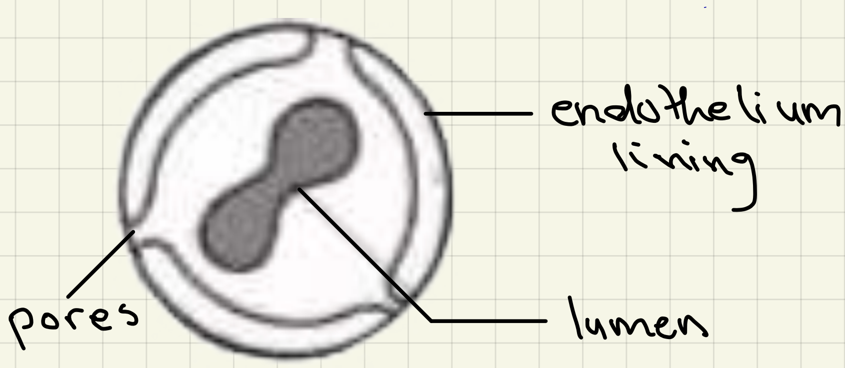
* carry deoxygenated blood back to heart except for:

pulmonary vein → oxygenated blood from lung to heart

umbilical vein → oxygenated blood from Placenta to fetus

C) capillaries:

- are one cell thick for more efficient diffusion.



3 Blood:

* Functions of blood:

① transport medium for materials needed for cells //carries away metabolic waste (by diffusion or active transport)

② Carrying hormones (chemical messages) from one part to the other.

③ provide immunity

④ Blood clotting

* Components of blood:

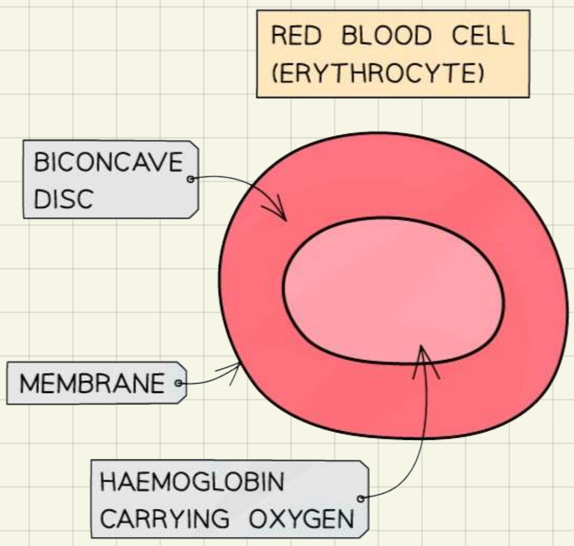
└ Solid → Red blood cells
  └ white blood cells
  └ platelets.
└ liquid → plasma
  └ water

→ Red Blood Cells (Erythrocytes):

• Main function: transport oxygen from lungs to cells.

• Adaptations: - contain haemoglobin; red pigment that carries oxygen and gives the cells their colour.
- Have a biconcave shape ; have a large surface area to volume ratio so oxygen diffuses in and out of cell rapidly
- Have no nucleus, more space for haemoglobin that carry oxygen
(Red blood cells have a life span of 120 days)
- smaller than white blood cells yet more in number

(Made in bone marrow)

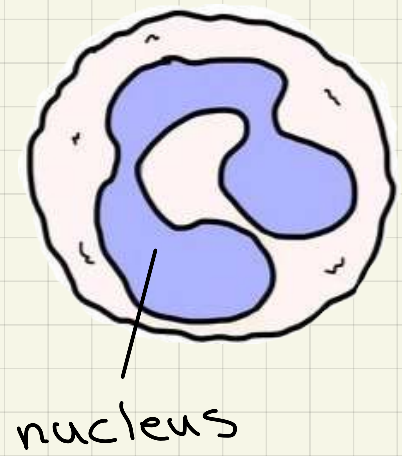


→ white blood cells (Leucocytes):

• Main function: Immunity

• Adaptation: Large yet can Squeeze through blood vessels

* important in inflammatory response when a tissue is damaged.
(Made in bone marrow mainly, sometimes in thymus gland)



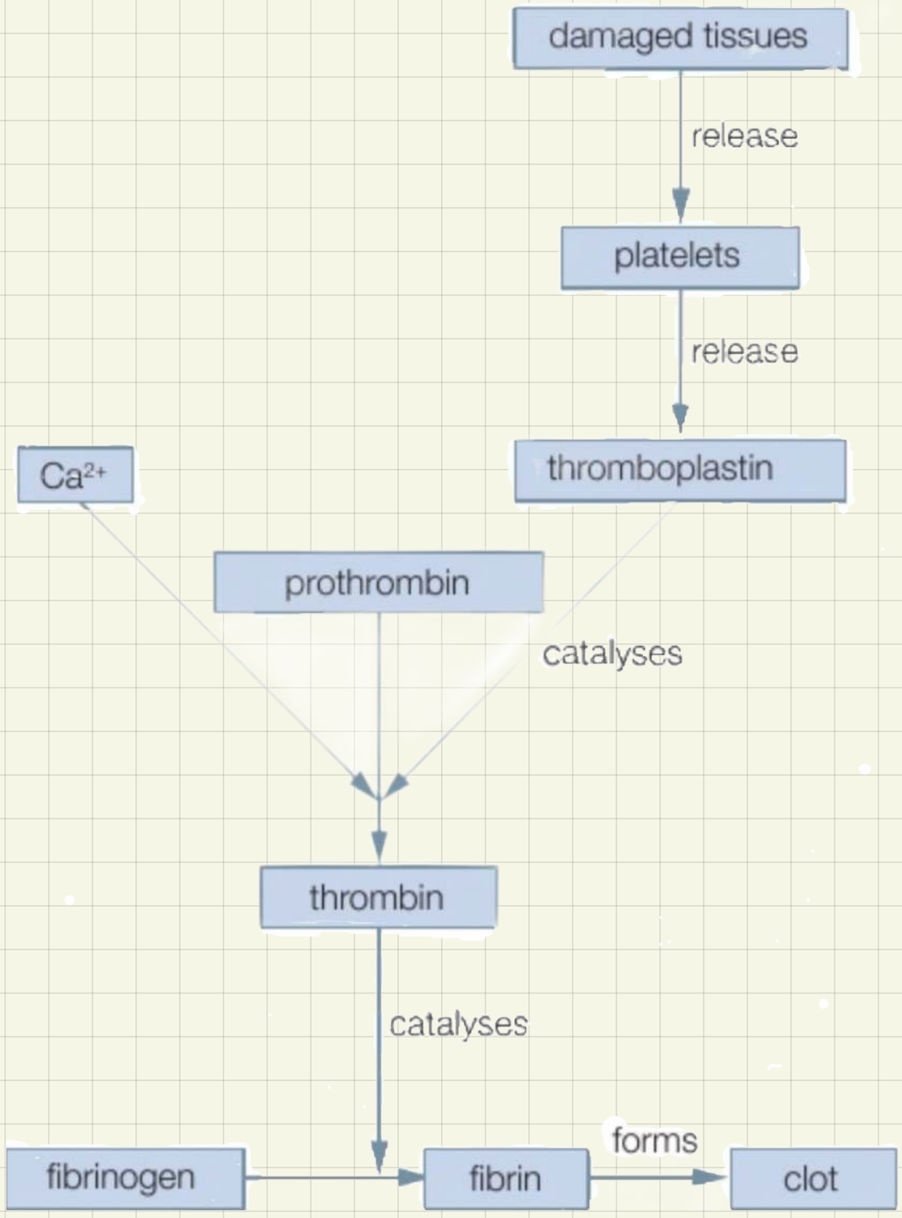
→ Platelets (Megakaryocytes)

• Main function: Blood dotting

↳ importance: i) prevents the entrance of pathogens.
ii) minimize blood loss.

↳ How does a blood clot form?

- serotonin; causes smooth vessel muscle to contract cutting blood flow to damaged area



- thromboplastin (active enzyme/soluble)
↳ catalyses the conversion of Prothrombin (plasma protein, unactive, soluble) to thrombin (active enzyme)
* calcium ions should be present at right concentration for this reaction to take place.

↳ thrombin active site binds to fibrinogen (soluble, plasma protein). to produce the product fibrin

⟹ Forming a mesh of fibre trapping blood cells and covering the wound forming a clot

* a special protein in the platelets structure contracts making the clot tighter and tougher to form a scab protecting skin and blood vessel as they heal

* cardio vascular diseases:

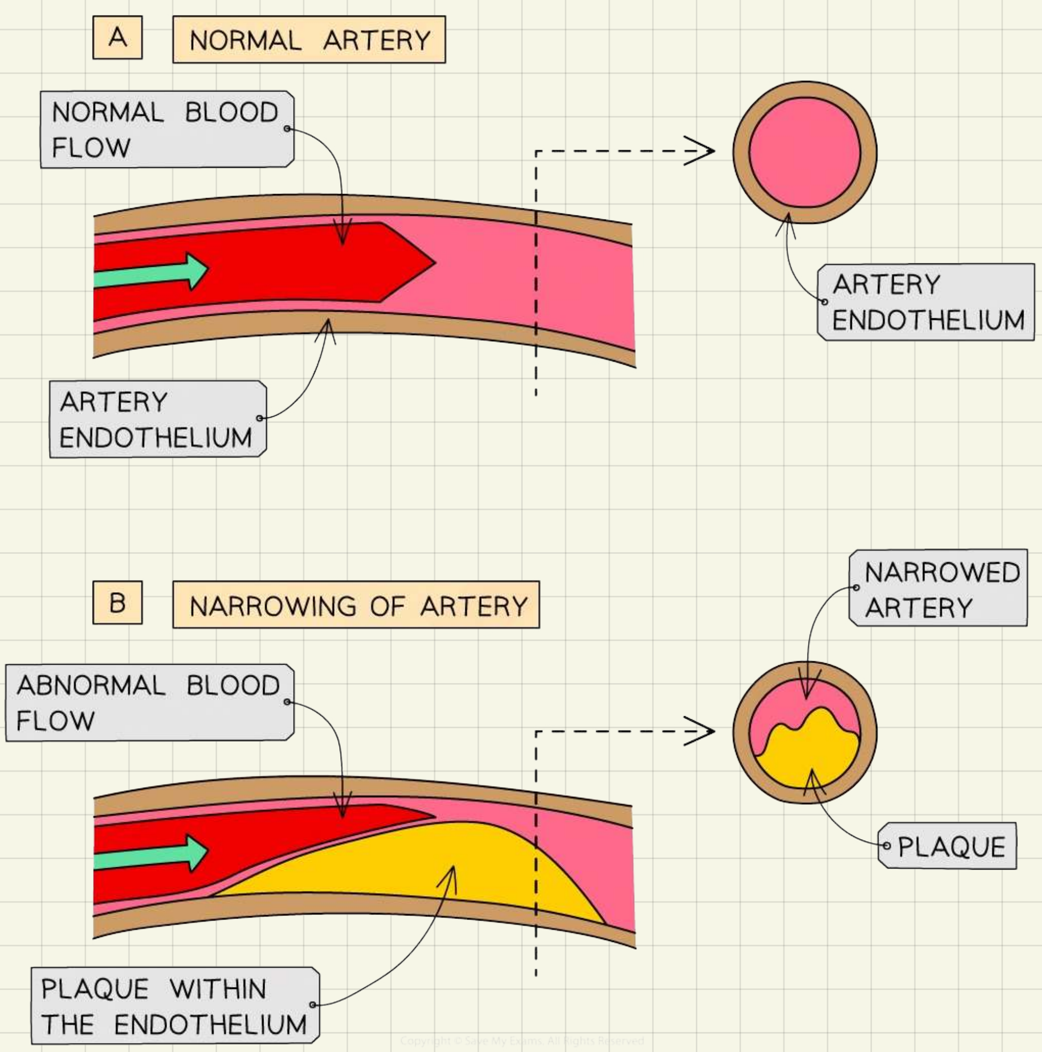
CVD:

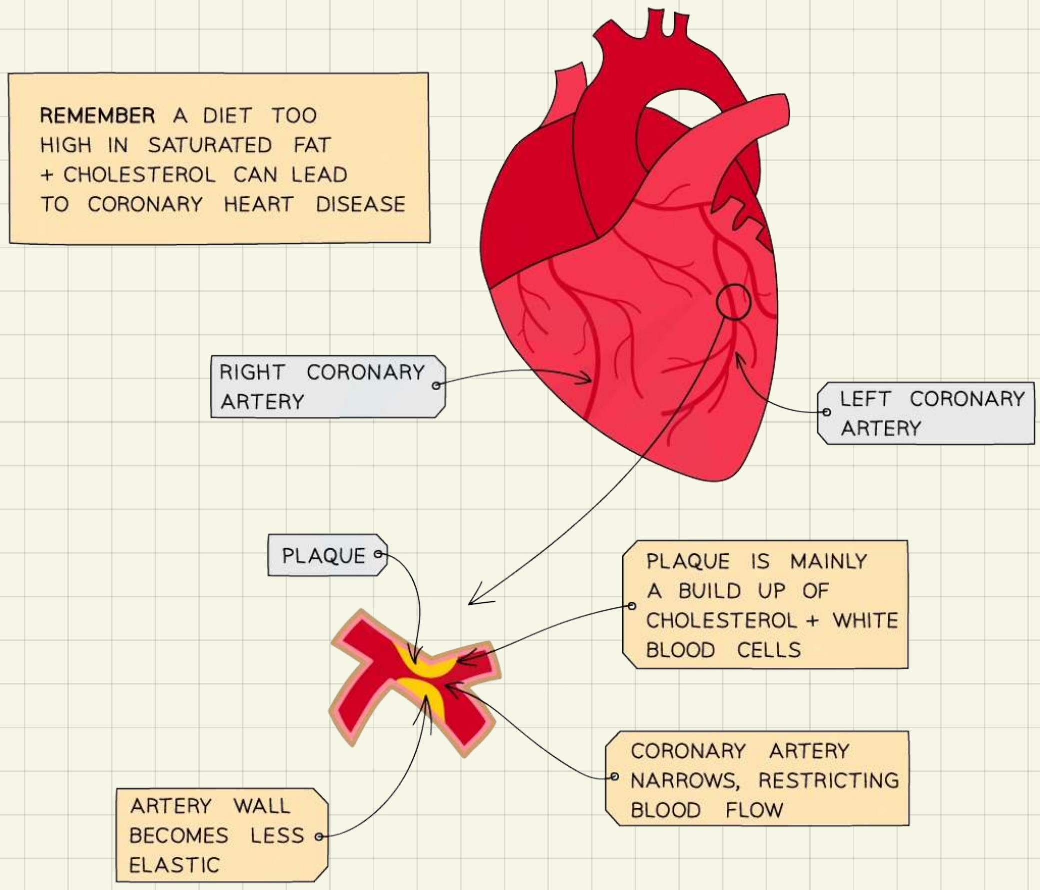
- Atherosclerosis:

↳ happens movingly (90%-95%) in arteries; as blood pressure in arteries is higher than in veins.

→ high blood pressure/smoking causes damage to the smooth endothelium lining of the vessel which stimulates an inflammatory response in which white blood cells accumulate and released cholesterol (a lipids) mainly and some other chemicals, so atheroma (plaque) forms and is surrounded by calcium and fibrous tissue resulting in the narrowing and handing of blood vessel; constricting blood flow to cells.

* when blood vessel narrows this increases already high blood pressure, increasing risk of damage in other areas of smooth endothelium lining.



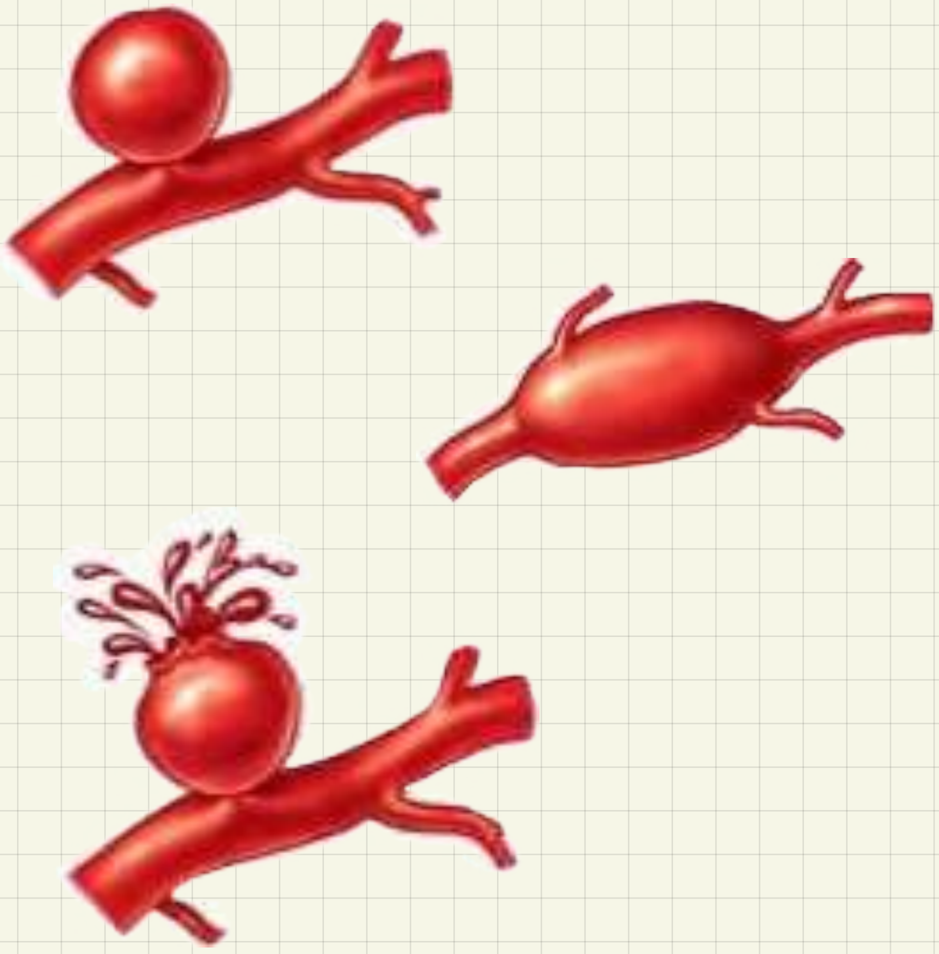


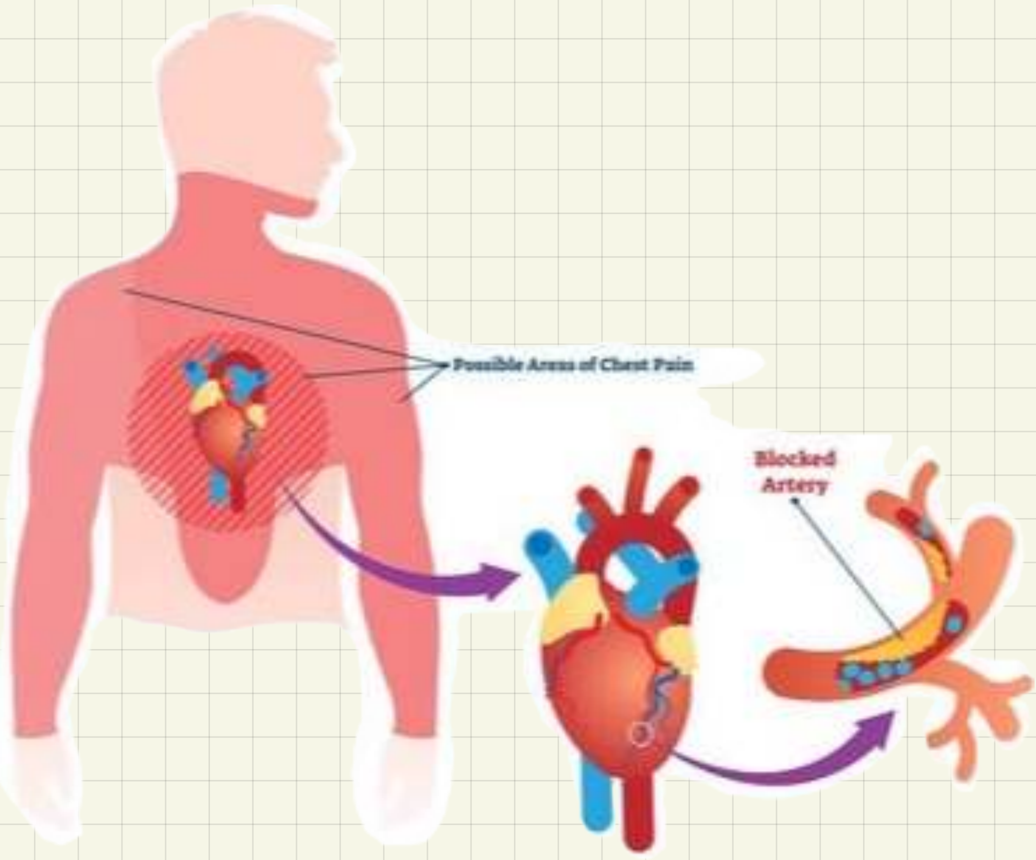
* it leads to:
- Increased blood pressure (kidney failure, blindness)
- Aneurysms, thinning of artery lining and internal bleeding
- leads to blood clotting (happens after atheroma)
- Heart Attak, if coronary artery is blocked
- Stroke if a brain supplying artery is blocked.

^ Normal blood pressure: 120/80

- Aneurysms:
when blood artery is narrowed by Atheroma blood tends to collect behind the blockage, this creates extra pressure at artery walls than normal causing it to get weekend. Weakened artery may split causing a massive internal bleeding

* usually happens in blood vessels supplying brain or aorta.
↳ the massive blood loss and drop in blood pressure is often faked
—> can be treated with surgery before vessel burst



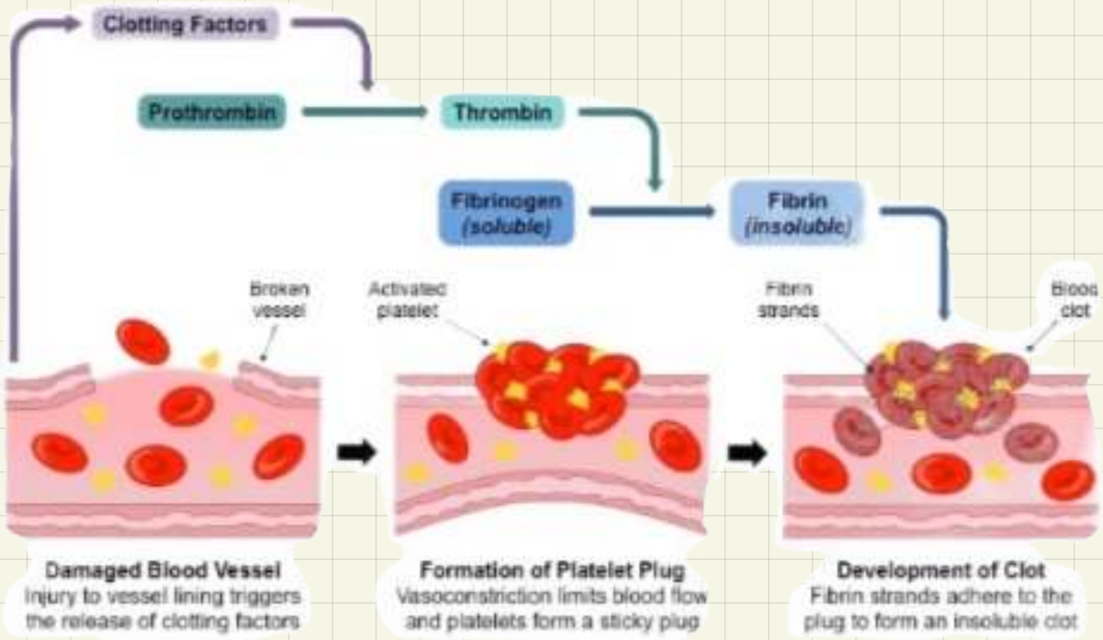


-Angina:
feeling pain at left side of chest after exercise due to narrowing in the coronary artery since there is less blood → less oxygen → less respiration → less energy, so energy is made by anaerobic respiration releasing lactic acid in heart tissues being supplied by artery wha!

* symptoms: pain in chest, arms (especially left), jaw/ breathlessness
* treatment: in mild conditions: exercise, losing weight, not smoking/ dilating blood vessel drugs

-Heart Attack:
feeling pain wait left side of chest without execrating, due to blockage in coronary artery so no * oxygen is delivered → no respiration → no energy → cells die.

* blockage is usually from blood clothing; as artery stiffened due to atherosclerosis it is easier to damage triggering platelets. this clot is called "thrombosis"



*Risk factor:
(possibility of something happening)

Uncontrolled

↳ age:
as age increases risk of CVD increases as arteries become less elastic thus harder and can't maintain high blood pressure

↳ gender:
less> 50 years old.
Males are at higher risk than females; as females blood flow has estrogen which maintains elastic lining of artery
Above> 50 years old.
Risk is the same in males and females

↳ genetics:
passed down naturally from parent to child through generations
(such as: high blood pressure / narrow blood artery)

Controlled

↳ smoking:
chemicals there damage lining of endothelium lining.
- high blood pressure due to carbon monoxide binding to haemoglobin so less oxygen is carried so more blood is pumped to meet cell oxygen demand.

↳ weight:
intake> required
→ obesity
(Body mass index= weight/height ^2)
Or (waist to hip ratio// cm)
when overweight blood is pumped more to reach extra body tissues
(blood pressure increases)

↳ diet:
-Cholesterol (narrowing of artery as cholesterol accumulate at artery walls)
- Salt (increase in blood pressure)
- Alcohol (increase in blood pressure)

* why do people take risks:
① private experience (people may be thin even while consuming high amounts of fatty foods)
② peer pressure
③ Addiction

* Antioxidants:

molecules that inhibit the oxidation of other molecules which can lead to chain reaction that may damage cells.

↳ such as vitamin (C):

an antioxidant that neutralizes free radicals

↳ From digested food

↳ cause cancer/ damage cells

↳ such as endothelial cells/ causing atherosclerosis

vitamin C is important for the formation of connective tissues in the body, so an increase in dietary vitamin C could help to reduce the damage to the artery endothelial layer that leads to atherosclerosis

* Investigation ① :

Detection of vitamin (c) :

→ independent variable:
- species
- storage time
- storage temperature
- Age

→ dependent variable;
- vitamin (C) concentration measured using DCPIP dye (changes from blue to colourless)

→ controlled variable:
- time
- temperature
- age
- other Independent variable.

* take results → repeat → mean → reliable results

* investigate to show the effect of storage time on vitamin (C) concentration.

-Get several fruit from the same species and age.

- place all fruit at the same temperature

- at different storage time, squeeze fruit to get it's juice

- Then by titration, add the juice drop by drop to a known volume and concentration of DCPIP dye.

- Find volume of juice needed to change the colour of DCPIP dye from blue to colourless.

- use calibration curve to find vitamin (c) concentration

- repeat the investigation multiply times and take mean for more reliable test results

correlation and causation:

correlation: strong tendency for two sets of data to change together
causation: when a factor directly causes a specific effect.
—> high blood pressure causes atherosclerosis but shows a correlation with heart attacks. Correlation ≠ Causation
[multi factorial diseases; needs several factors to occur, such as: cancer]

* A Good study design should:

- have a large sample site.
- longitudinal study
- From all around the world
- Control of most/all variables
- precise
- standardization

✰ treatment of cardio vascular disease:

① hypertensives: -used to treat high blood pressure.-

→ diuretics: increases volume of urine produced; so blood volume decreases; decreasing blood pressure

⇒ increased urine volume means excess fluids and salts are eliminated

→ beta blockers: block response of heart to hormones (adrenaline)

⇒ blocks heart receptors as a result adrenaline can't bind to it so, heart rate slows and contractions are weaker; blood pressure decreases.

→ Symphatic nerve inhibitors: stops impulse from reaching heart vessel so vasoconstriction does not happen.

⇒ affect sympathetic nerves; which causes arteries to constrict, running from CNS to rest of body. So arteries are dilated and blood pressure is low.

→ angiotensin inhibitors (ACE inhibitors): blocks production of angiotensin, hormone which stimulates constriction of arteries, keeping blood pressure low.

* they also reduce risk of CVD and kidney/eye damage due to high blood pressure

• side effects:

- o sudden drop in blood pressure
- o coughing
- o swelling of ankles.
- o tiredness and fatigue
- o constipation

*Antihypertensives affect the quality of life; as high blood pressure doesn't have any true symptom. Many patients leave medication as the side effects make them ignore the much larger but invisible risk of CVD.

② statins: - used to treat high cholesterol levels -

↳ block enzyme in liver that is responsible for making cholesterol
↳ decrease production of LDLs
↳ improve balance of LDL:HDL

* reduce inflation in lining of arteries.

- side effects:
  - o kidney and liver damage
  - o nausea
  - o muscle inflammation
  - o dizziness

* Plant stands and sterols: reduce amount of cholesterol absorbed by gut; have similar structure to cholesterol are sold in yoghurts, so it is easier on body to metabolize cholesterol and reduce levels ot LDLs in blood.
~ No scientific reason for why they work but there is evidence that they are affective.

③ medication to ' prevent blood clotting:

i) Anticoagulants: interfere with production of prothrombin
⇒ lower concentration of prothrombin to convert fibrinogen to fibrin, this increase blood clotting time. [example: warfarin]

ii) platelets inhibitors: make platelets less sticky
⇒ reduces clot ability of blood. [example: aspirin]

- Side effects:
  - o internal bleeding
  - o causes irritation of stomach lining (aspirin)

*Molecules, transport and health:

→ Inorganic ions:

- ↳ Nitrate ions: needed in plants to make DNA and amino acids, therefore responsible for marking proteins from photosynthesis's products

- ↳ Phosphate ions: needed to make ATP and ADP + DNA and RNA

- ↳ Chloride ion: needed in nerve impulses,sweating and many secretory systems in animals.

- ↳ Hydrogen carbonate ions: needed to buffer blood pH to prevent it from becoming too acidic.

- All of above are anions, negatively charged molecules.

- ↳ Sodium ions: needed nerve impulses,sweating and many secretory systems in animals.

- ↳ Calcium ions: needed for middle labella in plants + formation of bones and muscle contraction in animals.

- ↳ Hydrogen ions: cellular respiration and photosynthesis +pH balance

- ↳ Magnesium ion: needed for production of chlorophyll in plants

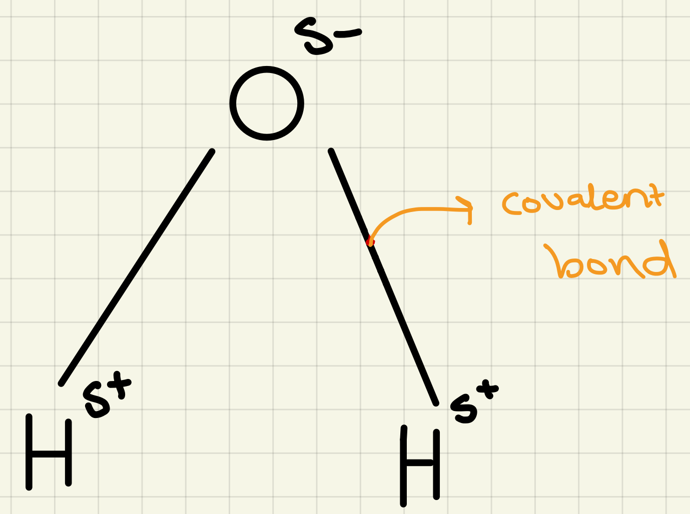
- all above ions are cations, positively charged molecules

✩ Water

- Polar substance → good solvent, it dissolves all ionic substances.

- covelent bonds, slightly charged so his a polar molecule and oxygen is dipolar

- covelent bond are easy to break. But hydrogen bonds are strong; holding water molecules much stronger



⇒ functions of water :

→ good transport medium

→ good solvent

→ habitat for many organisms (due to its high heat capacity, since hydrogen bonds need a lot of energy to break them, it is slow to absorb heat and release heat)

→ H2O molecules are adhesives they are attracted to different molecules; this helps inplant transport system

→ H2O molecules are cohesive so water molecules stick together; this helps in movement of water from roots to leaves.

→ water has a very high surface tension since attraction between hydrogen bonds is greater than attraction between water molecules with air creating a skin surface tension; this helps in plant transport system and water masses ecosystems.

Solubility
↓
Evenly distributed without being broken down

Dissolve
↓
Breaks down and disappears in solvent

* Hydrogen bonds in water is what explains the properties of water

Carbohydrates

↳ Monosaccharides:
- sweet
- easily dissolves in water

→ Examples:
~glucose $(CH_2N)_n$
~fructose
~galactose

↳ Disaccharides:
- sweet
- dissolves in water

→ Examples:

~ sucrose < Glucose, Fructose

~ lactose < Glucose, Galactose

~ maltose < Glucose, Glucose

↳ Polysaccharides:
- insoluble in water

→ Examples:
~ starch — in plants
~ glycogen — when body had excess glucose it converts to glycogen for storage
~ cellulose — in plants cell wall structure

- Glucose:

1) alpha glucose

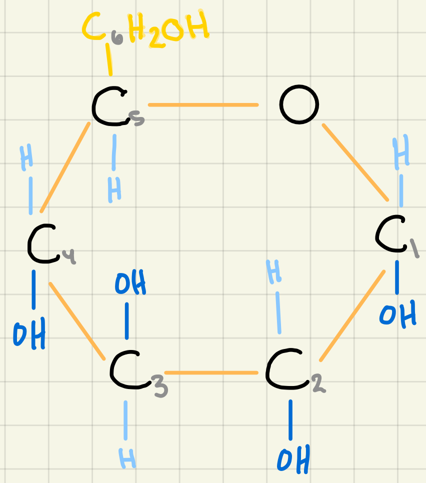
α

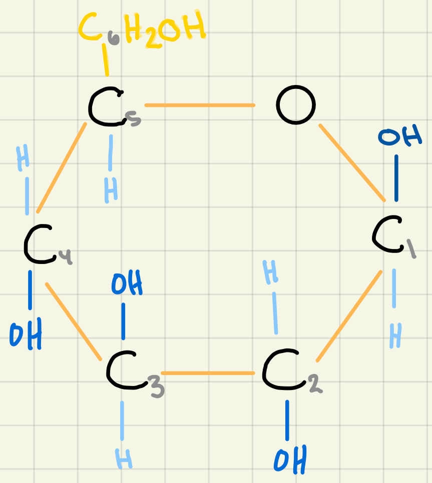
2) beta close

β

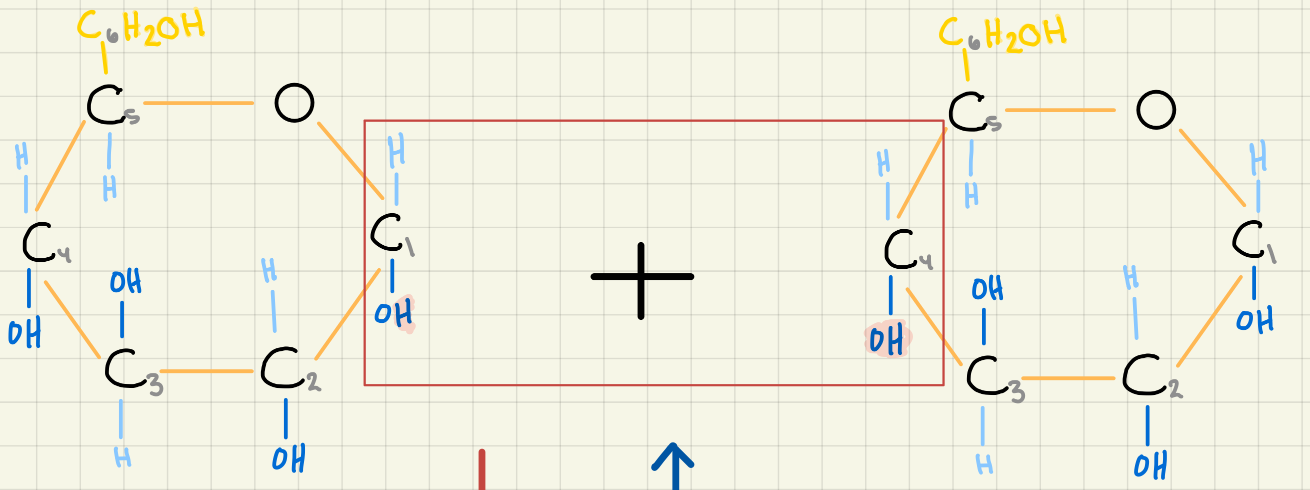
↓ Structure

↓ Structure



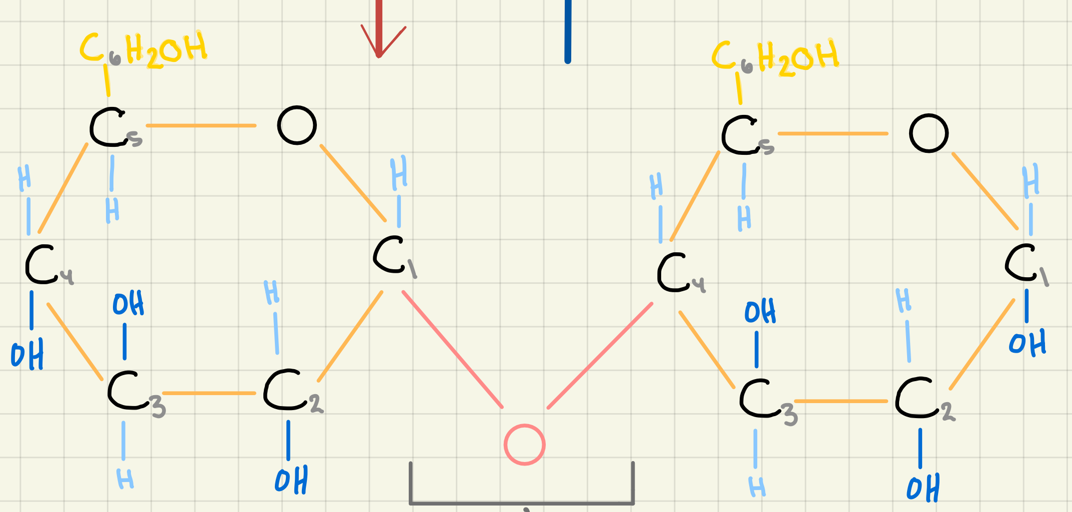


- Disaccharide: maltose → glucose + glucose.



Condensation: removing a water molecule

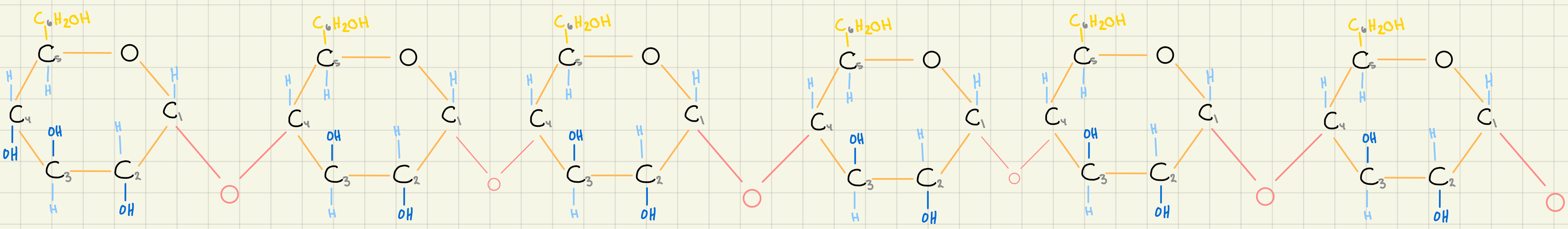
Hydrolysis: adding of water (splits the bond)



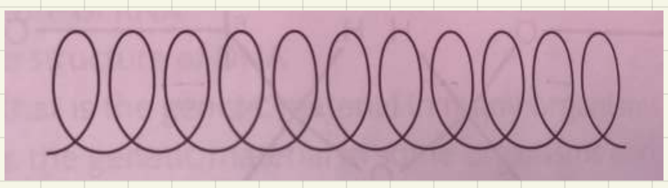
(1,4)
glycosidic bond

- **Polysaccharide:**

① Amylose

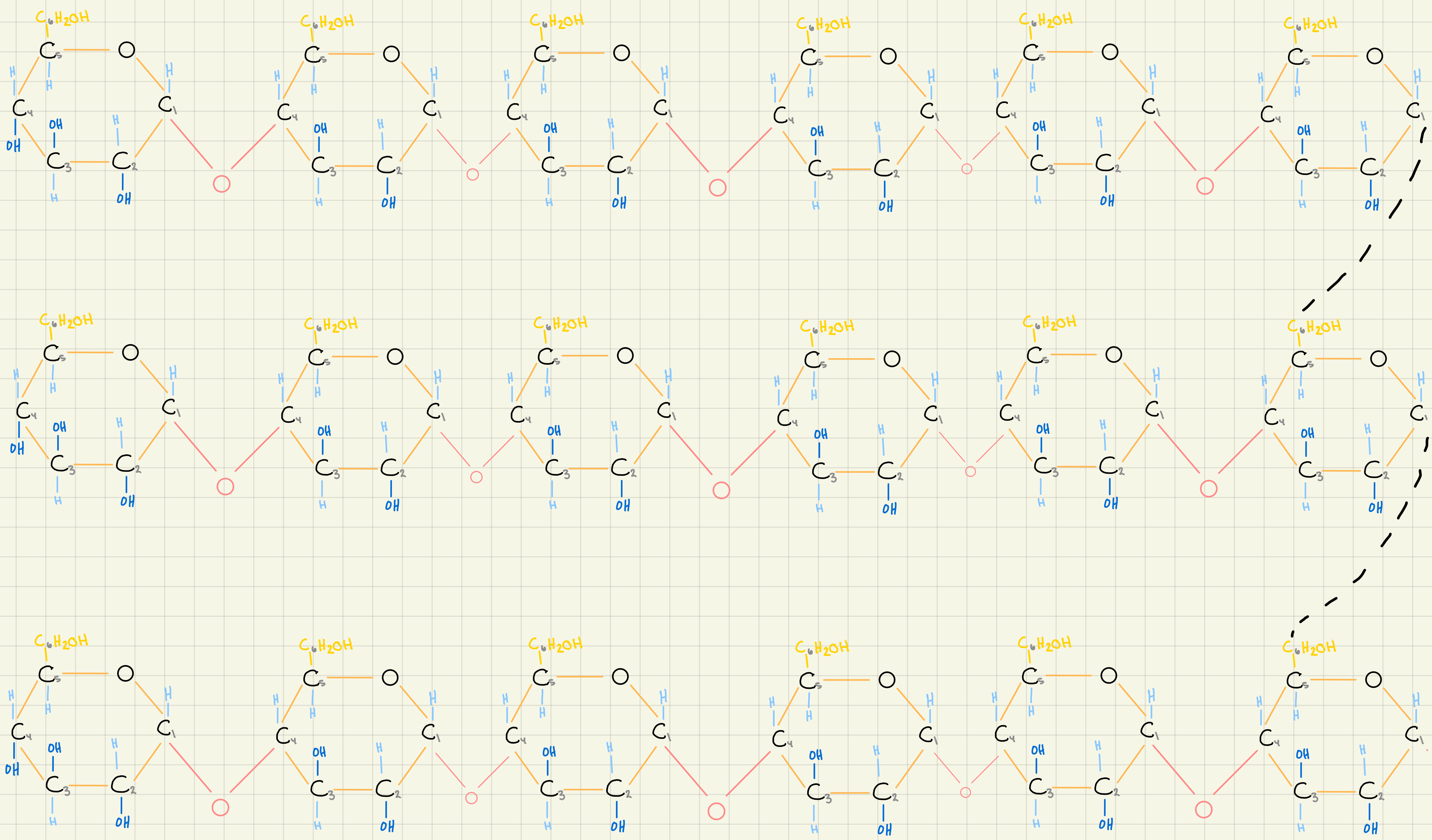


Has only (1,4) glycosidic bond
not branched
Chain of repeating α glucose



Amylose chain forms a spiral.

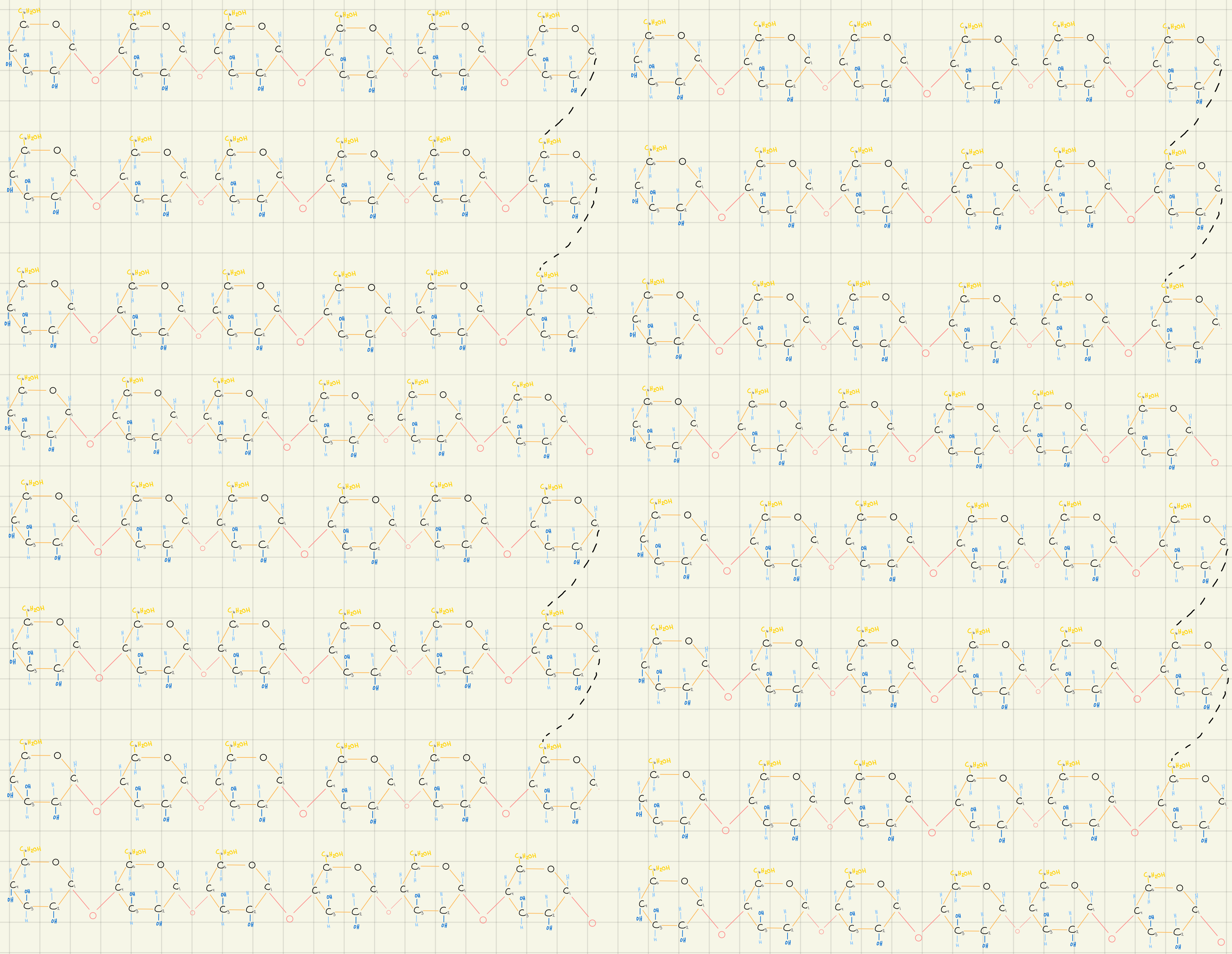
② Amylopectin



Has both (1,4) glycosidic bond and (1,6) glycosidic bond.
Branched so they have many glucose terminals
Provides more energy when split up

③ Starch

- Energy store in plants (not source)
- insoluble and compact
- can be rapidly broken to release glucose to release energy
- is a mixture of amylose and amylopectin
- is a polysaccharide// polymer of α glucose



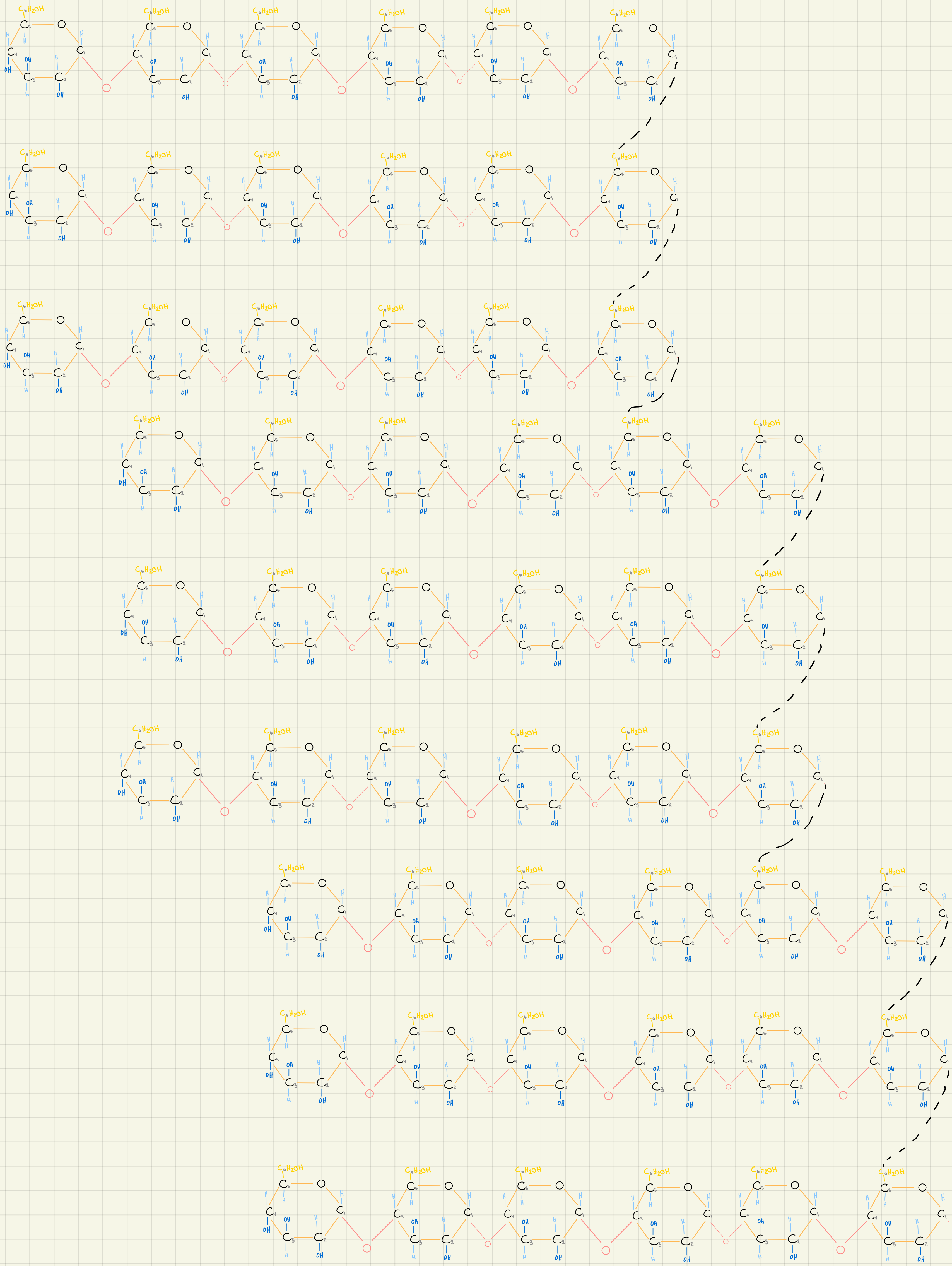
* starch is broken done to glucose by hydrolysis, to be used directly to release energy in respiration.

All polysaccharides are chemically and physically inactive so they don't interfere with other cell functions

④ glycogen

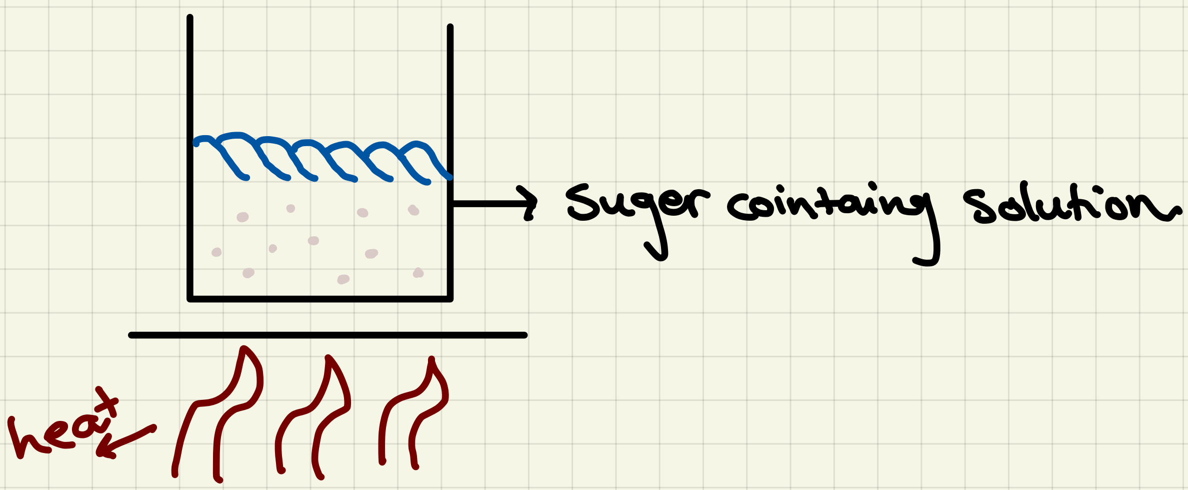
- Excess glucose in humans are stored as glycogen
- Is a polysaccharide// polymer of Glucose
- Is branched with both (1,4) and (1,6) glycosidic bond
- Has more branches than amylopectin

—>As a result has more terminals of glucose so it releases more energy



Measurement of reducing sugars:

→ Benedict reagent (original color: blue)



Color change from blue to:
Green
Yellow
Orange
Brick red
(The more intense/the faster the color changes the more Suger it has)

- all monosaccharides and disaccharides (except sucrose) are reducing sugars.

** you can heat other types of sugars with few drops of HCL, then cool it and add sodium hydrogen carbonate to neutralize it, this breaks the glycosidic bonds (by hydrolysis) producing a monosaccharides that can react with Benedict reagent.

Testing for starch:

Add a few drops of iodine solution (original color: reddish brown) to food sample wether solid or in a solution. If color changes to blue black then sample contains starch.

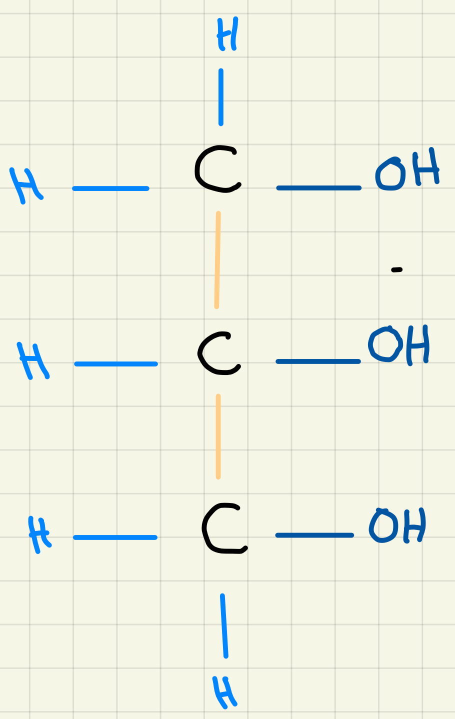
Lipids

Fats
(solid at room temperature)

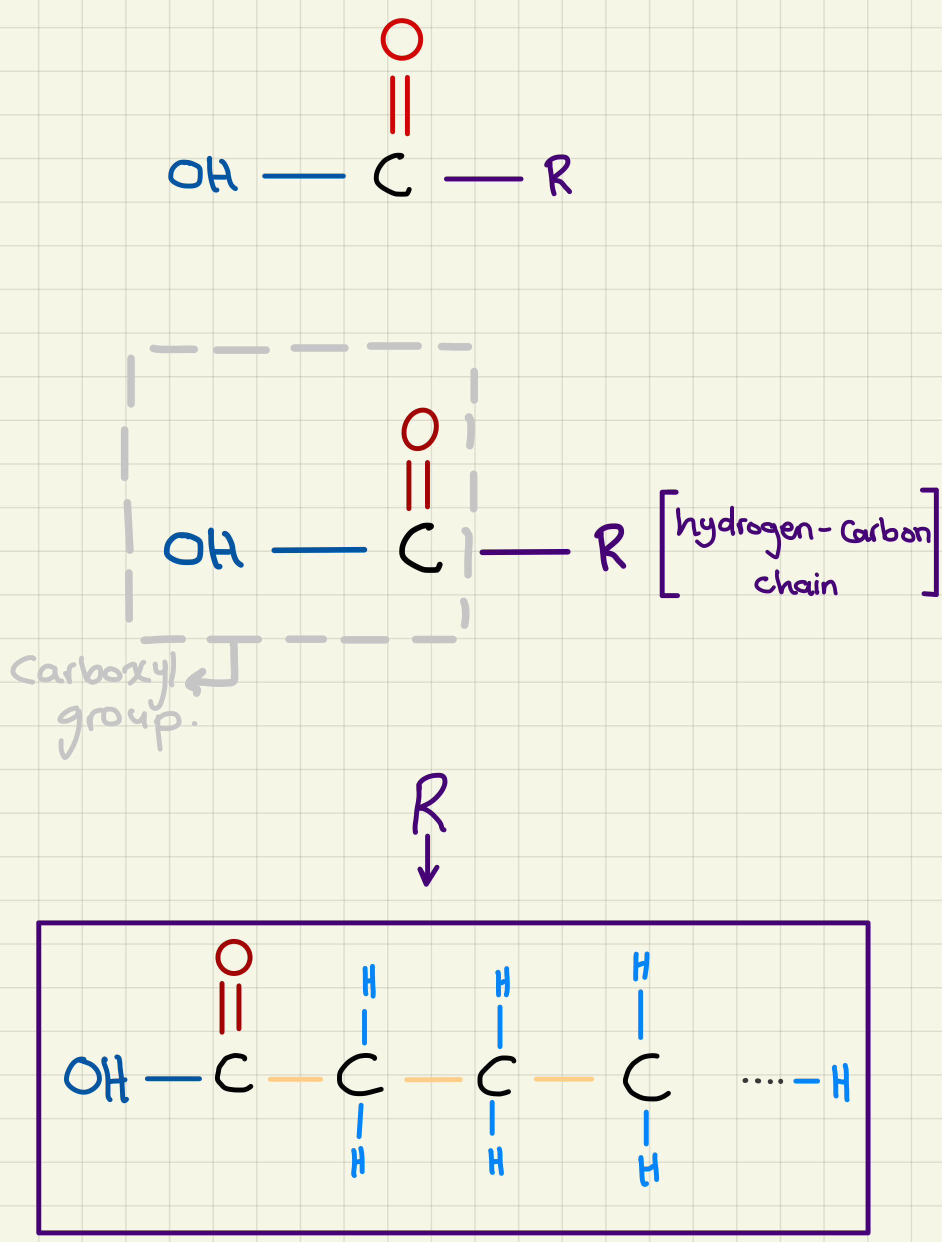
Oils
(liquid at room temperature)

⟹ Both structures are glycerol + fatty acid

Glycerol



Fatty acid



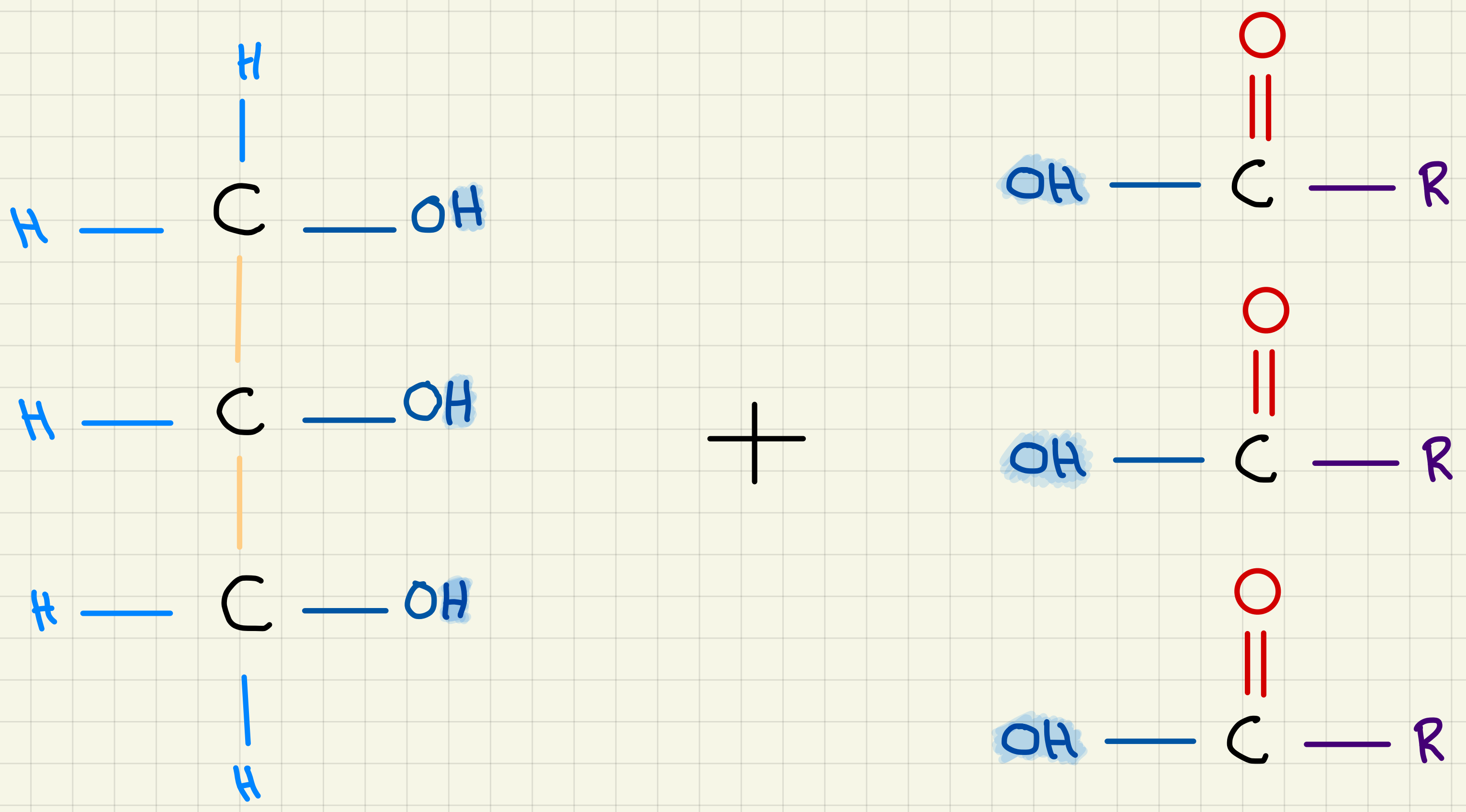
* Fatty acids is what changes the type of the lipid

The change can either be:

1) increase or decrease in the number of fatty acids found in a lipid
2) change in the fatty acid's R- group.

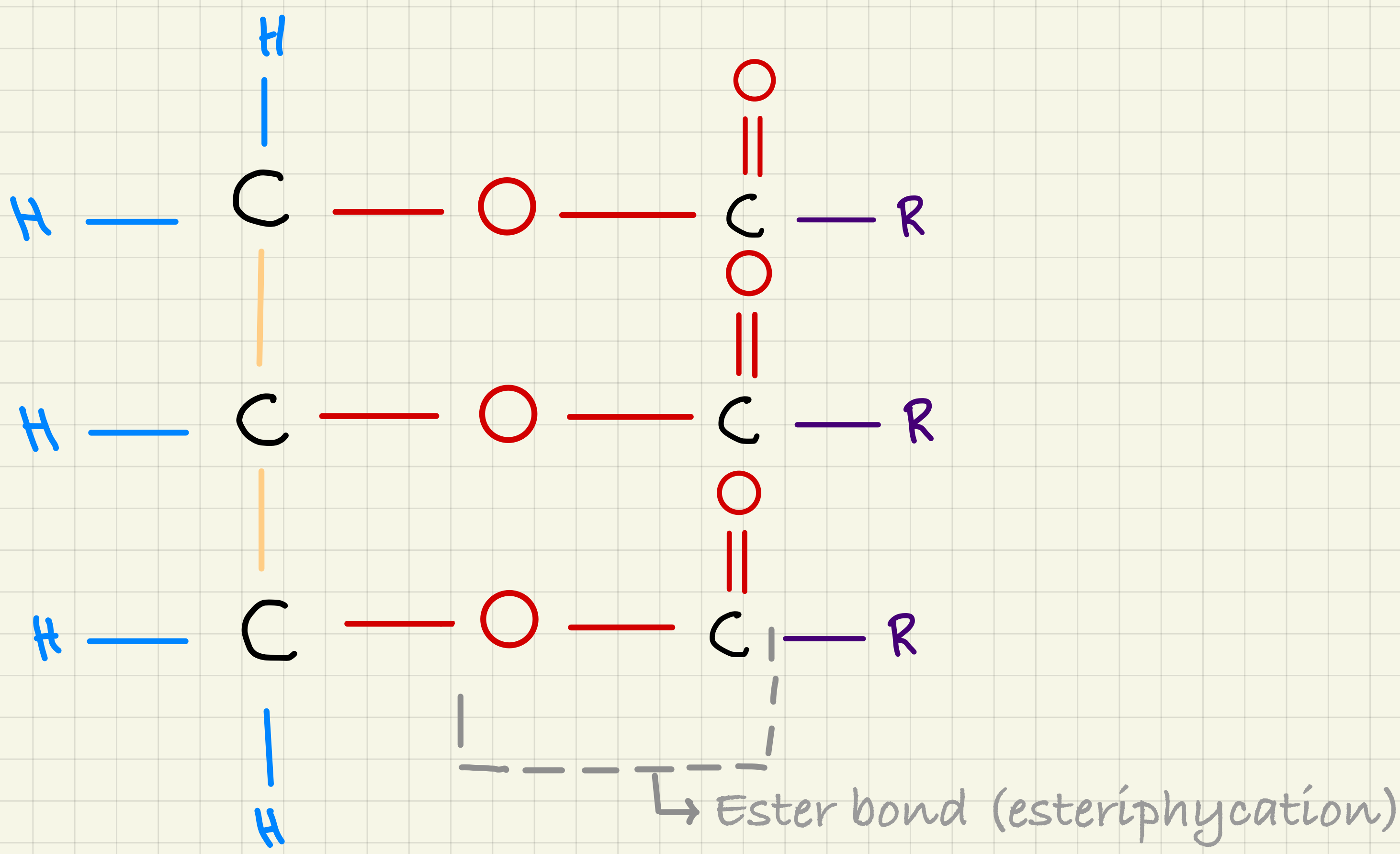
↳ Number of carbon in a R-group can be [15-17] or [16-18]

* Triglyceride: 1 glycerol + 3 fatty acid

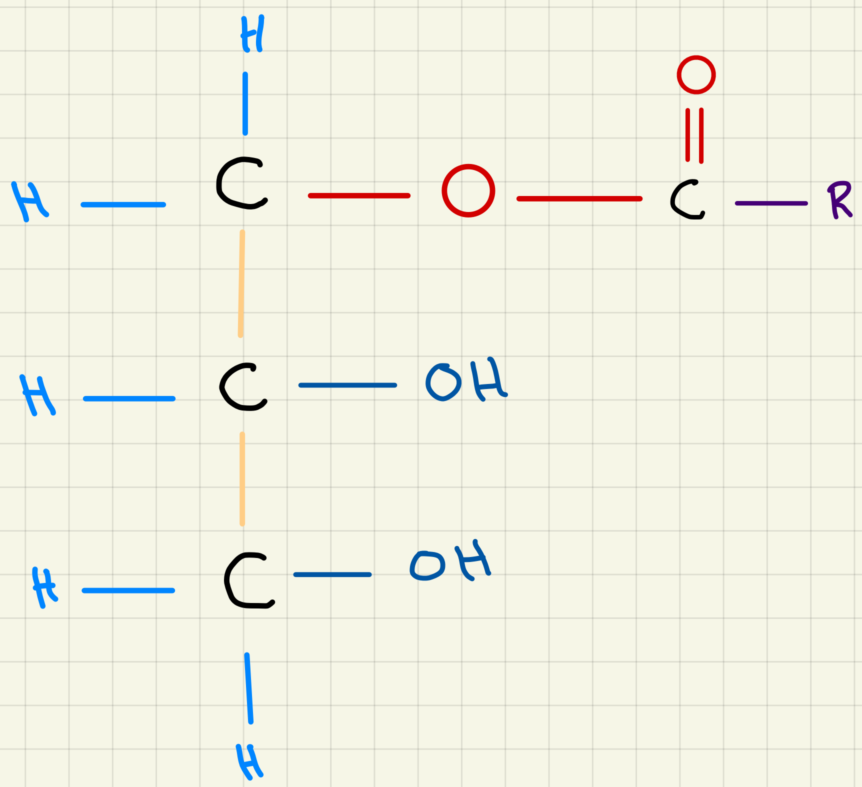


Hydrolysis ↑ ↓ condensation [gain of 3 H_2O molecules]

H+ OH = H2O

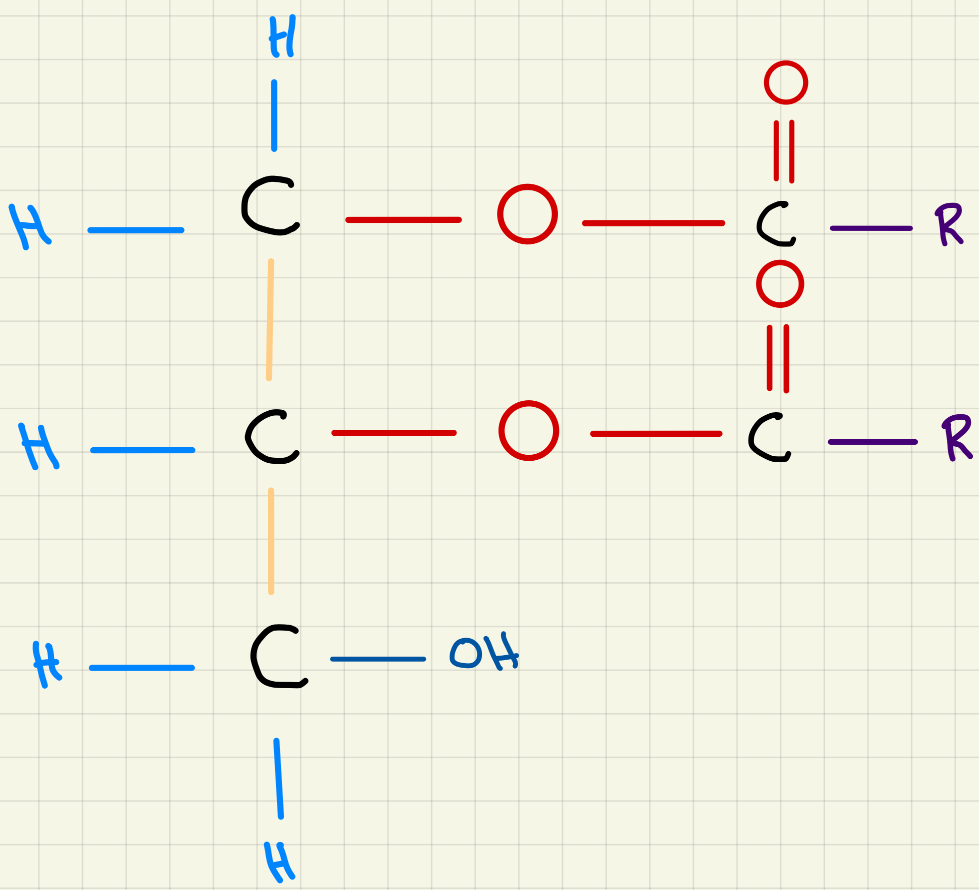


* one glycerol + fatty acid



$+ \; 1 H_2O$

* one glycerol + 2 fatty acids



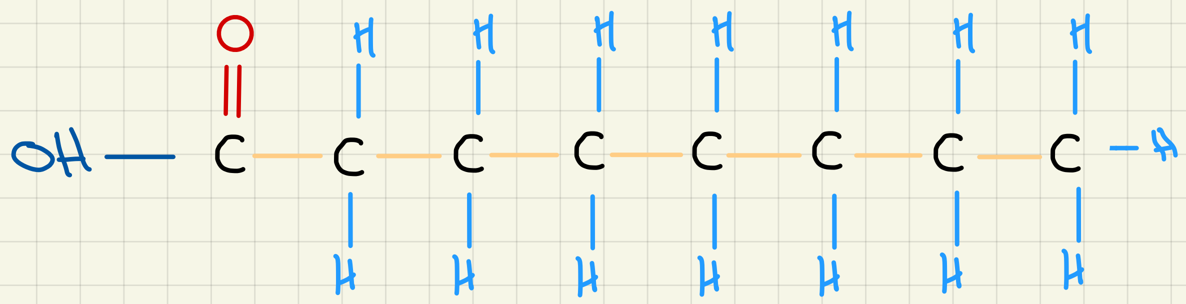
$+ \; 2 H_2O$

Maximum bond is only with 3 fatty acids, as there are only 3 carbons and [OH]s to bond with

Saturated fats:

Example: fats

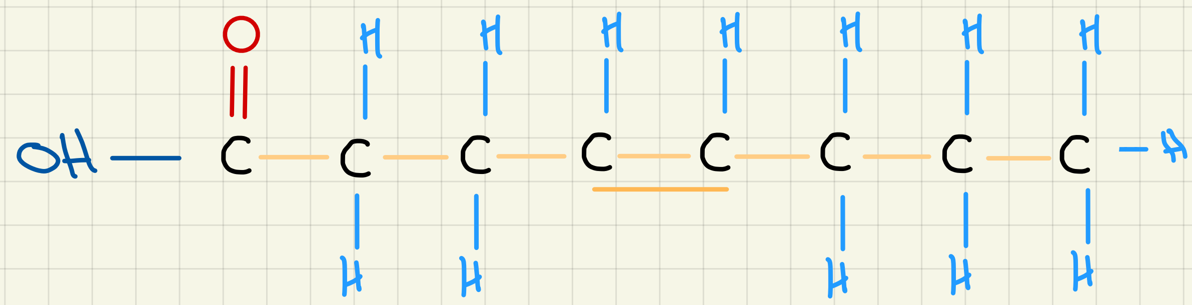
- they increase risk of CVD; since they can be easily deposited in artery
- there is no double bond between carbons
- carbon to hydrogen ratio is lower than unsaturated (more Hydrogen)



Unsaturated fats:

Example: oils

- slide through artery/ doesn't deposit easily
- decreased risk of CVD
- there is a double bond between carbons
- carbon to hydrogen ratio is higher (less hydrogen)



Saturated	-Have carboxyl group -Has a higher melting and boiling points than unsaturated -No double bonds -has higher carbon to hydrogen ratio
Unsaturated	-has a kink shape (tilt/bend) that makes it weaker/ easier to break - has a lower melting and boiling point since it is easier to break bonds between carbon and carbon than to break bonds between hydrogen and carbon - has double bonds

→ Relationship between lipids and CVD:

- Lipid → non-polar → not soluble in blood

So, it is turned in blood in form of lipoprotein.

↳ Lipid + conjugated protein which transport cholesterol

① High density lipoproteins (HDL):
Protein is higher in number than cholesterol
Has unsaturated lipids
Is "good cholesterol" ; since it stores cholesterol in cells so no risk of deposit in artery

② Low density lipoprotein (LDL):
Protein number is lower than cholesterol
Has saturated lipids
Is "bad cholesterol"; since it easily deposits in arteries + mores cholesterol from tissues to artery

The greatest risk of CVD is connected to ratio of LDL: HDL not lipid intake.
Many factors are taken into account when seeing LDL and HDL levels, such as: genetic makeup, metabolic rate.

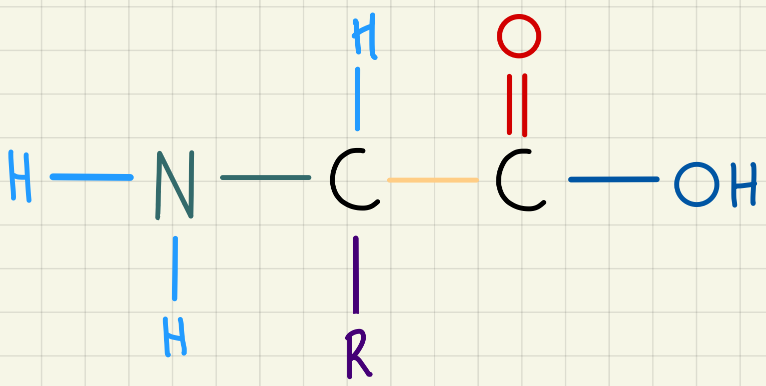
- Importance of lipids:
  a) insulation
  b) cell surface membrane
  c) source of energy (releases energy more than carbohydrates)
  d) insulation of neurons

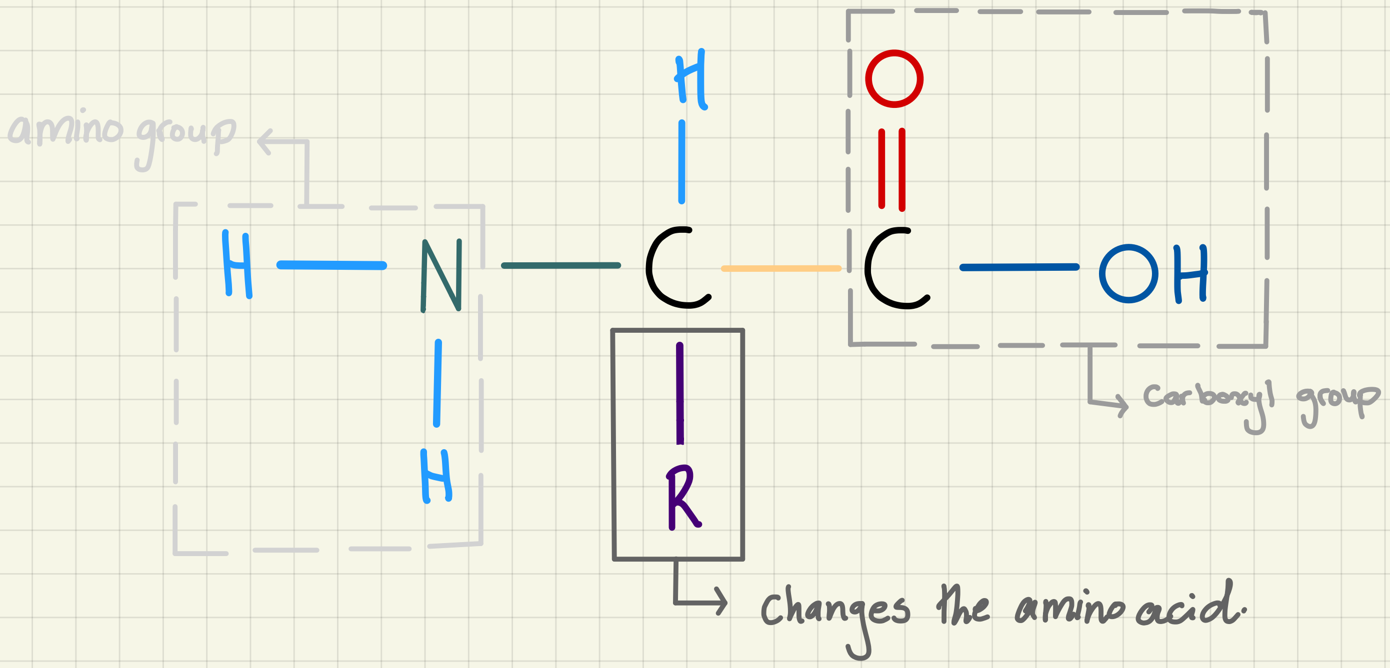
Protein:

* Importance:
  a) making enzymes
  b) forming cell membranes
  c) producing antibodies
  d) transport oxygen in form of heamoglobin

Proteins → amino acids → carbon, hydrogen oxygen, nitrogen (sometimes sulfur)

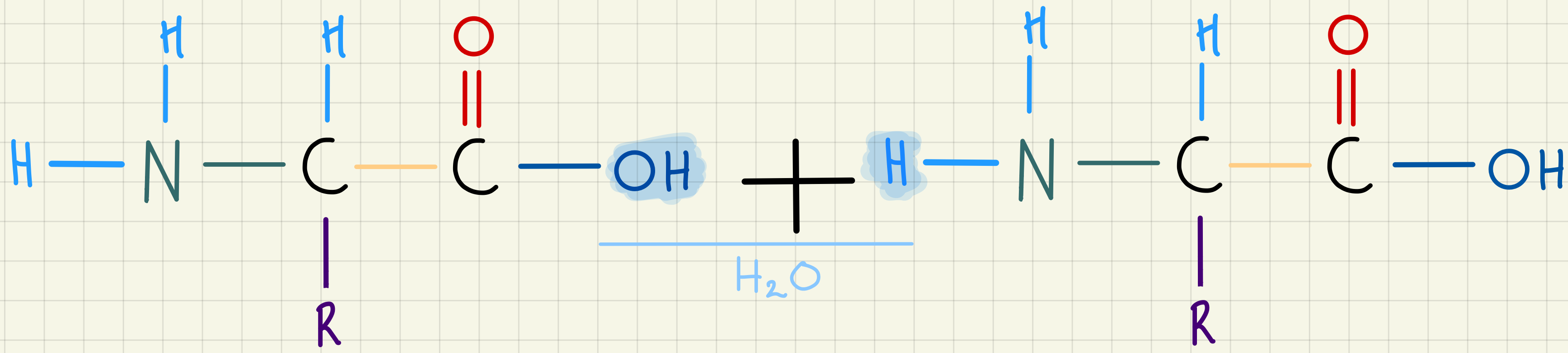
* general structure of amino acids:





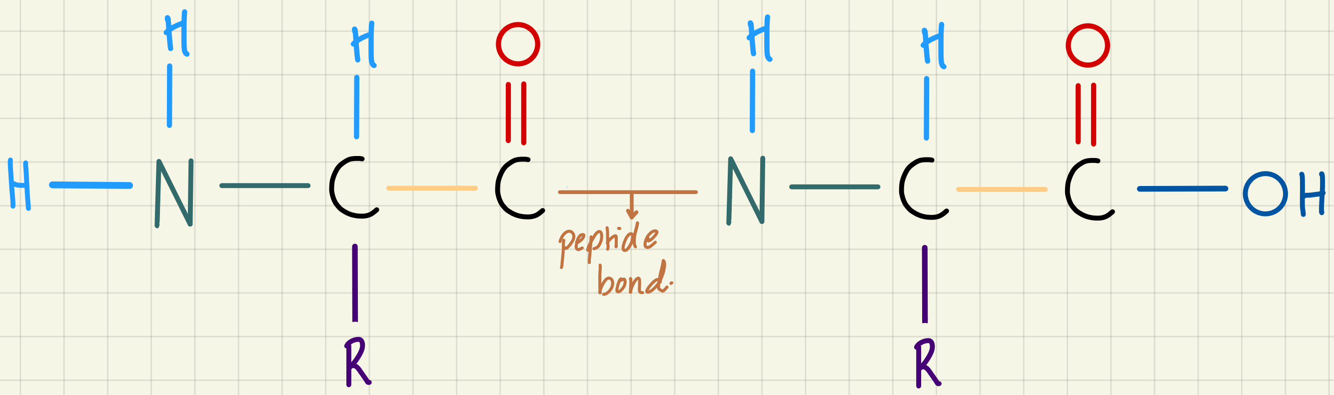
- Proteins differ due to different amino acid sequences, which determines type and position of R-group. Types of R-group determine types of bonds [ionic bonds]. Types of bonds determines folding into 3D-structure

⟹ Condensation and hydrolysis of amino acids:



Condensation: removing a water molecule

Hydrolysis: adding of water (splits the bond)



- The reaction occurs between carbonyl group and amino group
- the products for condensation are: dipeptide + H_2O (lost)
- the products for hydrolysis are: 2 amino acids + H_2O (gained)

Structures of protein:

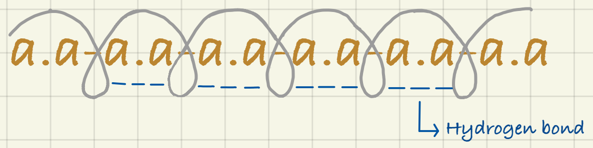
Primary structure:

a.a-a.a-a.a-a.a-a.a-a.a

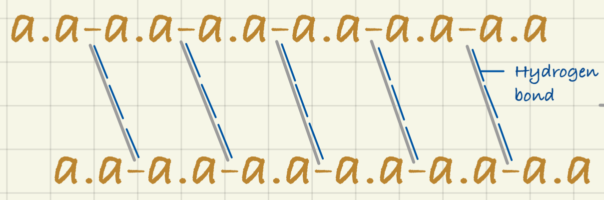
{only peptide bonds}

~ linear sequence of amino acids with only peptide bonds

Secondary structure:



→ Alpha helix (coil shape))

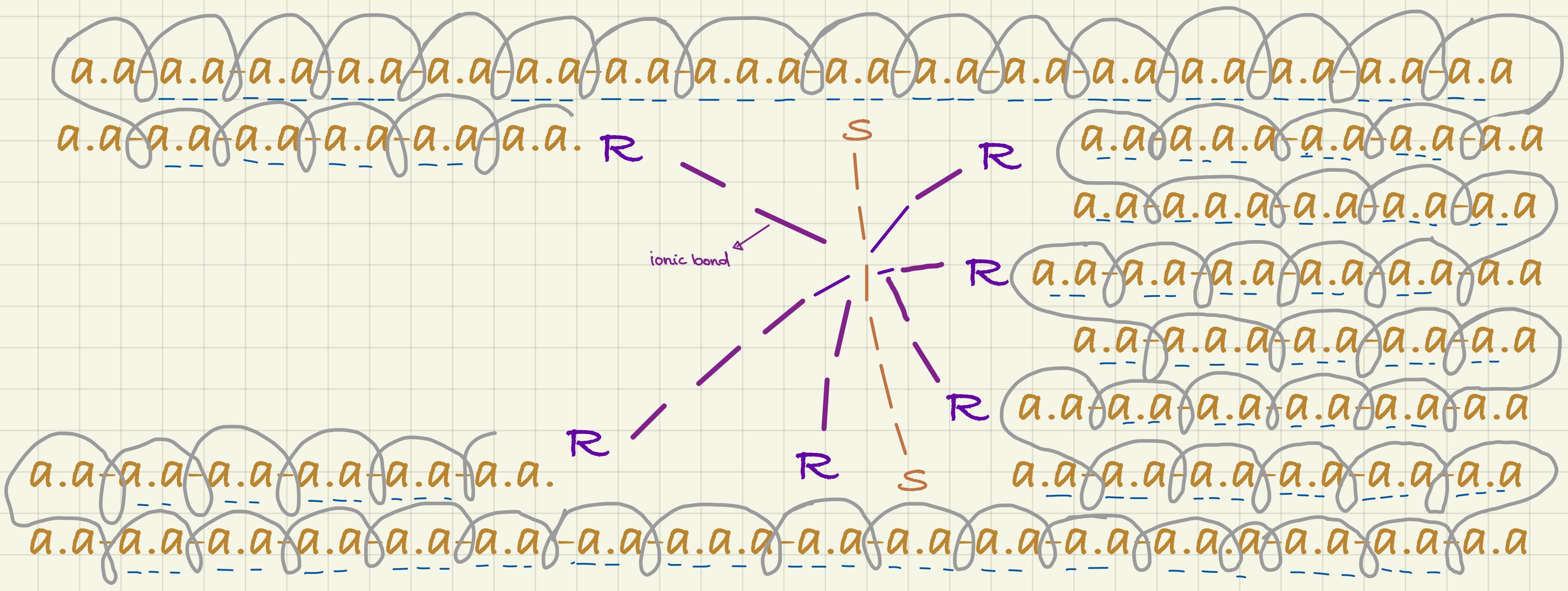


→ B-pleated sheets

{both peptide and hydrogen bond}

Tertiary structure:

The 3D folding of secondary structure



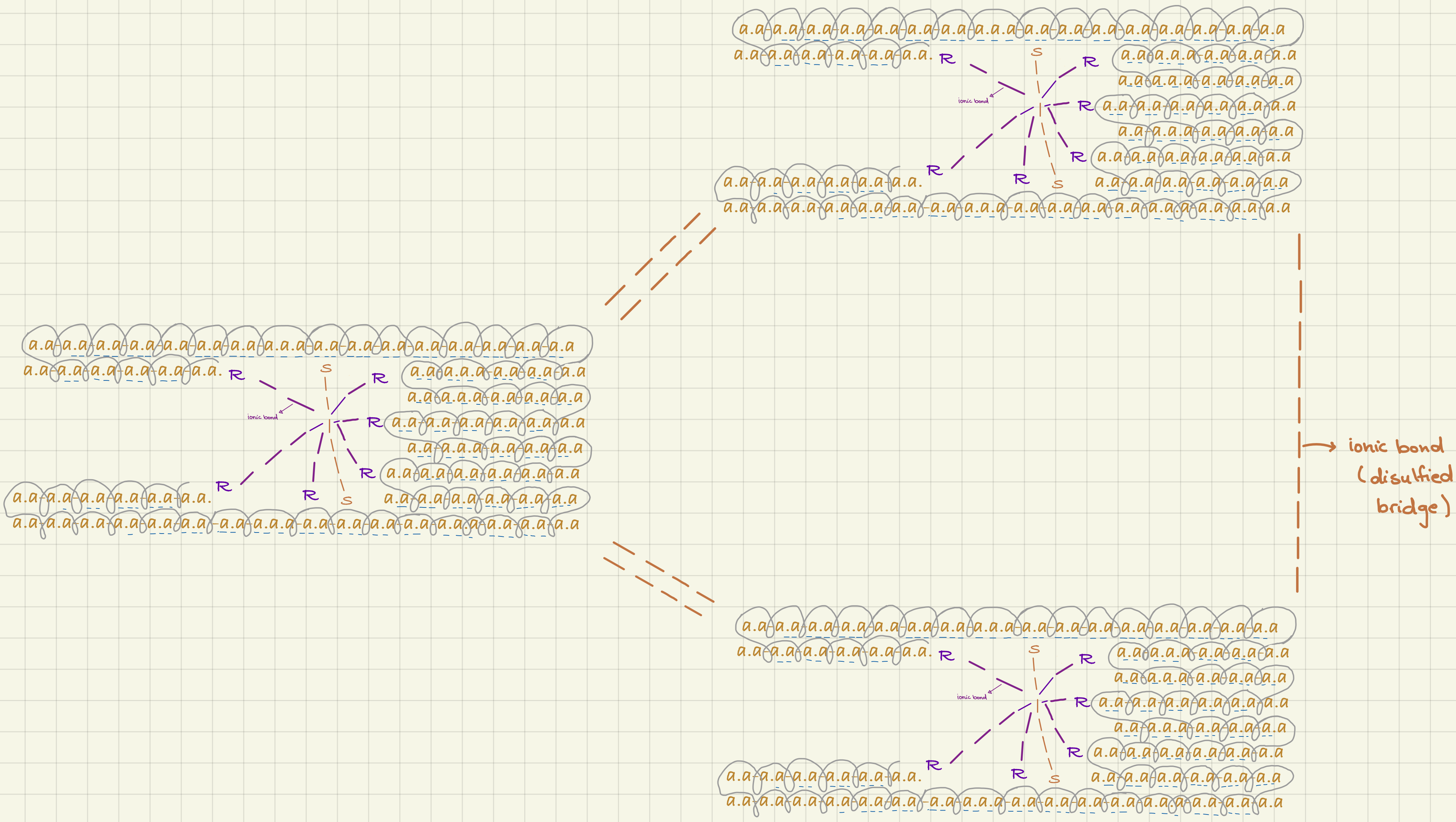
{ Hydrogen bond, peptide bond, ionic bond and disulfide bridge }

ionic bond → between R groups

disulfide bridge → between sulfur Atoms in a.a

Quaternary structure:

3D arrangement of more than one tertiary polypeptide



{ Hydrogen bond, peptide bond, ionic bond and disulfide bridge }

*Slight change in conditions, such as increase of temperature or pH can affect / break the bond which will result in loss of the 3D shape of protein.
[denaturation]

Types of proteins:

1) fibrous proteins:

-Long, parallel Polypeptide chains
-Have little to no tertiary structure
-Insoluble in water
-Have structural functions (found in connective tissues/ keratin which makes up hair, nails and feathers/ silk of spider webs)
- give strength to the structure their found in

Example of a fibrous proteins : collagen

-Found in tendons, ligaments, bones, skin to give strength
-Made up mostly from quaternary structure
-Has triple helix shape made from α chains which have an amino acid sequence of [glycine, proline, hydroxyprolin]
-The structure is held by hydrogen bonds
- encourages absorption of calcium in bones

2) globular protein:

- round/ sphere shape
- mostly made from tertiary structure
- soluble as they have hydrophilic R groups on the outside and hydrophobic R groups in the inside
- carboxyl and amino group end gives globular proteins ionic properties so it doesn't settle in water, and it is hard to separate from water.

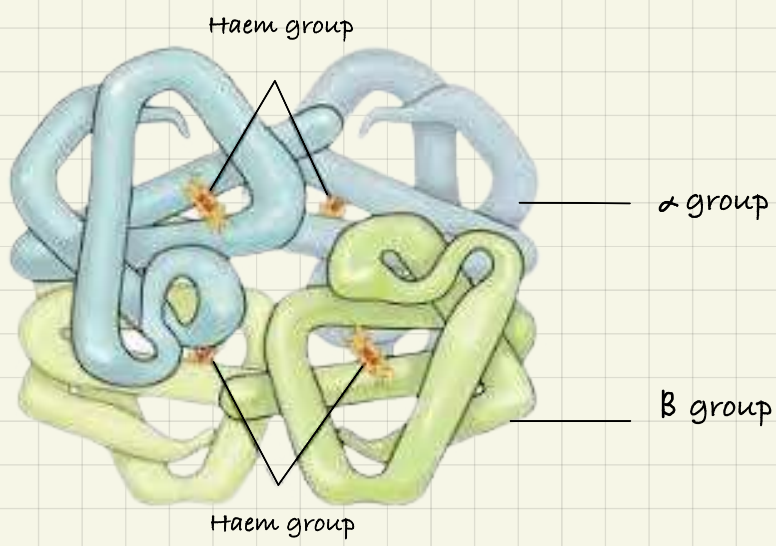
Example of a globular protein is: haemoglobin

⇒ Haemoglobin:

Haem ↓ Iron

Globin
- 2 α Polypeptide chain
- 2 β Polypeptide chain

(4 irons per molecules)
Each iron carries 1 O_2 molecules



3) conjugated proteins: (Sub type of globular)
protein + prosthetic group
↳ enhances the properties of protein

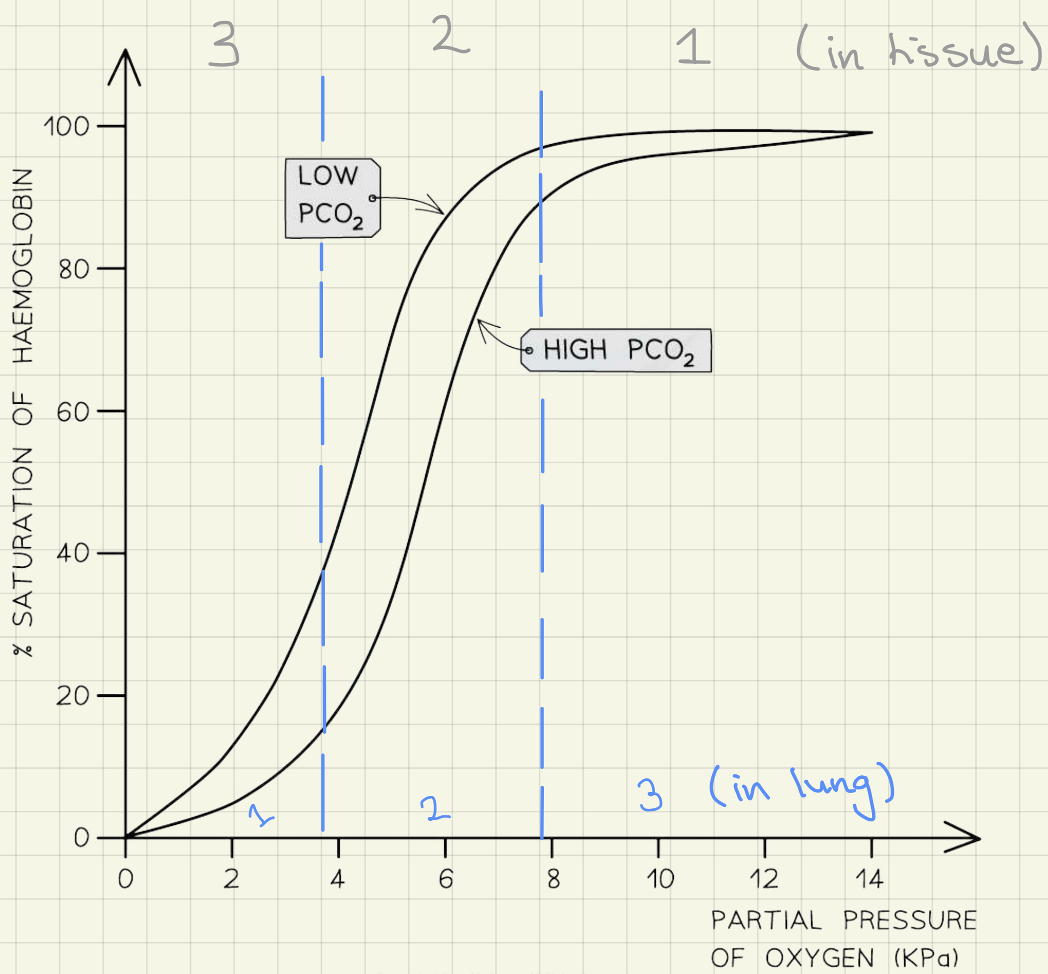
Examples of conjugated proteins:
a) haemoglobin
⟶ Makes protein able to carry oxygen
b) glycoprotein
⟶ Helps protein to become watery (mucus)
c) lipoprotein
⟶ Improves protein ability to transport cholesterol

~All types of protein contain all four structures but in different percentages ~

Transport of oxygen

Haemglobin + oxygen ⇋ oxyhemoglobin ($Hb4O_2$)

Dissociation curve:



Stage 1:
first bending of oxygen is difficult but it changes the arrangement of the molecule so it becomes easier for oxygen to bind with haemoglobin; so the gradient of graph is small

Stage 2:
Binding with haemglobin is easier thus faster; so gradient is steep and sharp

Stage 3:
No oxygen binds to haemoglobin as Hb is fully bonded with oxygen and is saturated; so there is no increase and the line is flat (gradient zero)

* In tissues the opposite is going to happen, stage 1 becomes 3 and so on but instead of bind it will be release of oxygen

Graphs shifts to the left when Hb has a higher affinity to bind with oxygen.

→ BOHR EFFECT

When carbon dioxide levels are high Hb need to bind with more oxygen to become saturated as a result Hb releases oxygen more easily (in tissues)	When carbon dioxide levels are low Hb doesn't need many oxygens to become saturated so it binds with oxygen faster (in lungs)

**When CO2% increases in blood, blood pH decreases since CO2 makes blood acidic. Hb is a protein and is affected by pH so its shape/ structure changes slightly which makes it ability to release O2 increases since oxygen is needed for respiring cells. This makes curve in graph shifts to the right.

→ At high altitudes, there is less O2 in air, so Hb affinity to bindincreases. So curve shifts to the left

→ FETAL HAEMOGLOBIN

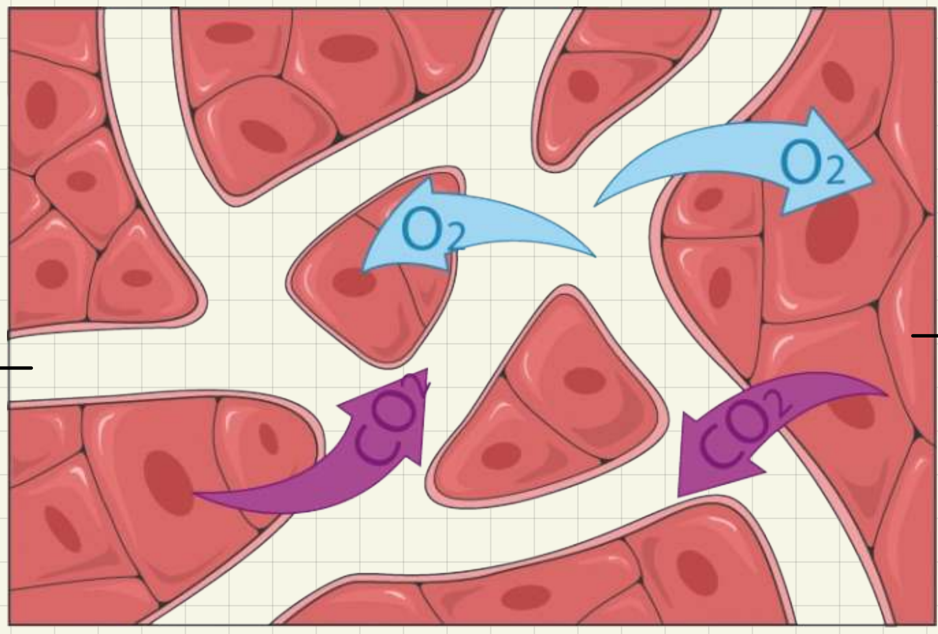
Contains 2 alpha and 2 gamma polypeptide chain

→ Has more affinity to bend with O2 since maternal blood has lower O2 concentration than air

**So the curve of fetal blood in graph shifts to the left; maximizing the oxygen intake from mother to fetus, since maternal blood contains less oxygen than oxygen gained from surroundings.

Transport of CO_2

High O_2 concentration from lungs (blood)



CO_2 concentration is high due to respiration (in tissue)

Carbon dioxide moves from cells to blood by diffusion

Most of CO_2 in blood dissolves in plasma

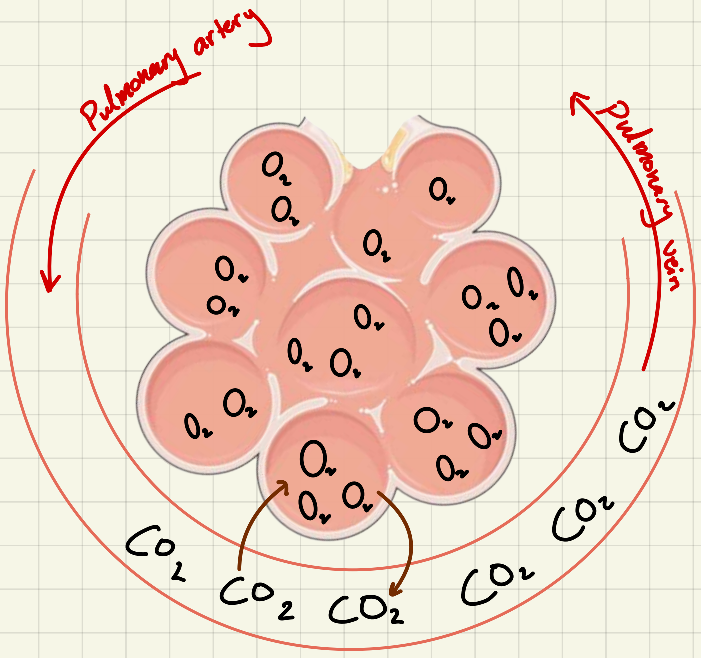
$\Rightarrow CO_2 + H_2O \leftrightharpoons H_2CO_3$ (acidic)

$\rightarrow H_2CO_3 \leftrightharpoons H^+ + HCO_3^-$ (acidic)

<u>The catalyst of this reaction is carbonic anhydrase (enzyme)</u>

Some of CO_2 binds with haemoglobin to make carbaminohaemoglobin

* concentrations of CO_2 in blood should stay low to keep a high rate of diffusion



In lungs, blood should have high concentration of CO_2 so diffusion gradient between capillary and alveoli is high and as a result fast, so reaction becomes

$\Rightarrow (H + HCO_3) \rightarrow H_2CO_3 \rightarrow CO_2 + H_2O$

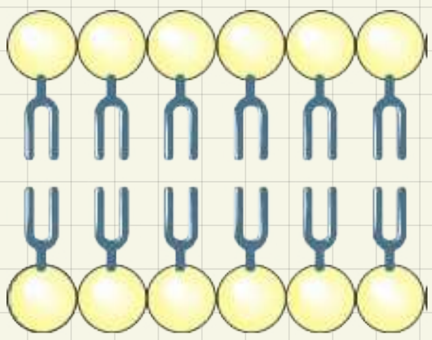
Unit 2: Membranes, protein, DNA and gene expression

Cell surface membrane:

↳ Made from: bilayer of phospholipid

↓ 2 layer

↓ - Phospho: phosphate group
- lipid: glycerol + 2 fatty acids

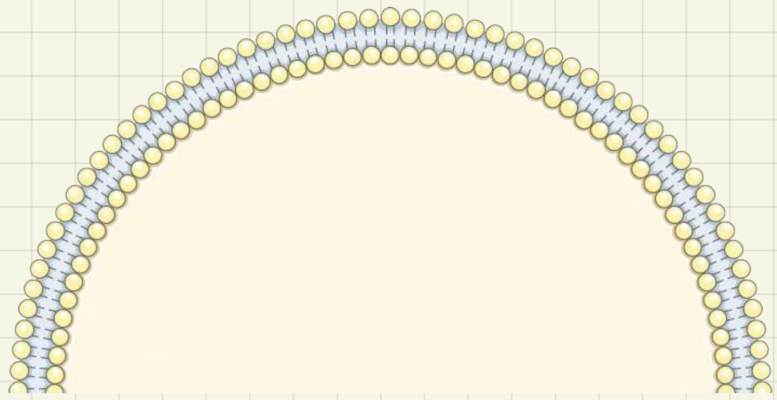


*Phosphate group are polar (charged)/ hydrophilic, can form hydrogen bonds with water and form a solution/ react with water

*Fatty acids are non-polar/ hydrophobic , do not reach with water, they repeal water

* Bilayer → since there is water both inside and outside the cell

[This structure is for all membranes]



⟹ Embedded In surface membrane:

1) protein carrier

-Can be found at surface of membrane (glycoprotein/ carbohydrate + protein) and acts as a specific molecule receptor

- Can be found on the internal cell membrane to control reactions linked to membrane

2) channel protein

- aids in active transport, needs energy, to transport material from high to low concentration
- pores/ gaps between channel proteins have hydrophilic R-groups which let polar substances move into/ out of the cell (facilitated diffusion)

3) cholesterol

Binds to fatty acidi limits movement of cell membrane (controls fluidly of the cell

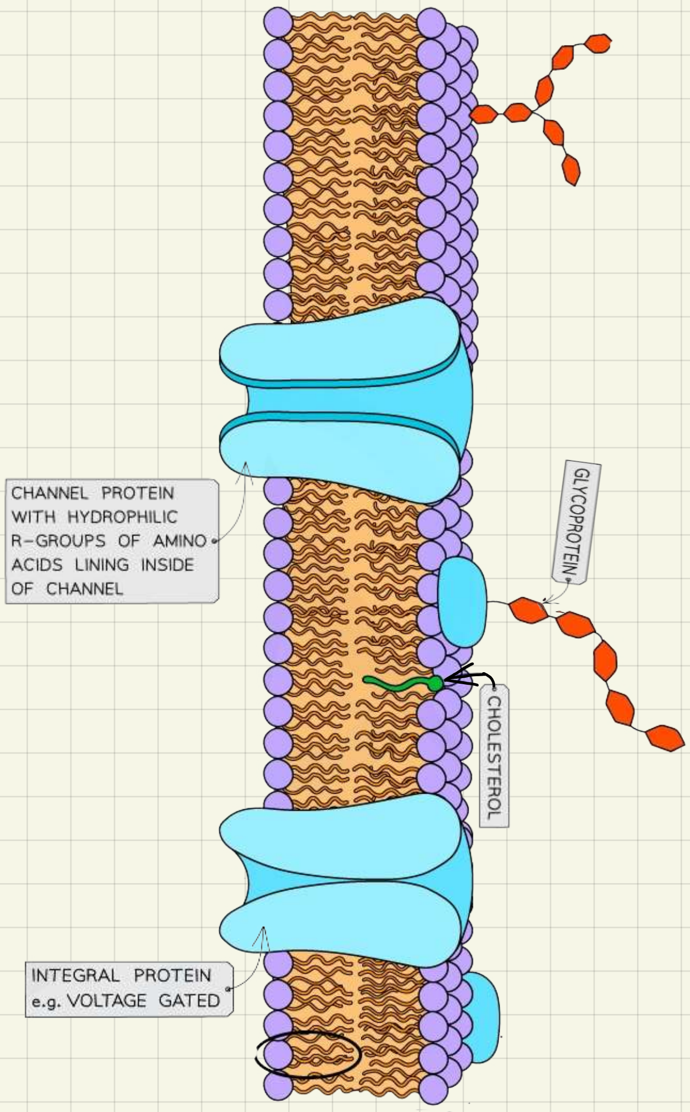
→ cell surface membrane is fluid,not rigid, flexible

* describe the structure of cell surface membrane :
Bilayer of phospholipid in which the phosphate head is hydrophilic (polar) and the fatty acid is hydro phobic (non-polar) there are different proteins embedded embedded in the phospholipid such as channel proteins, glycoprotein and cholesterol which binds to fatty acid tail

- Model of the membrane

Lipid soluble substance enter very easily —> most of membrane is lipid

When punctured membrane reseal them self —> membrane has fluid structure



- Hugh davson and James Danielle

(Made model of lipid center with protein on either side)

Treated membrane by propane to extract lipid, the two lines remained intact —> evidence that the two lines are proteins

X-ray showed pores —> distance between phosphate heads

Recent technology showed evidence for carrier proteins

*Why is the cell surface membrane is given the name of fluid mosaic? / meaning of the model fluid mosaic?
1)Due to the embedded protein in cell surface membrane.
2) it is fluid due to the random movement of fatty acid tail.

*Why is a bilayer of phospholipid is formed?
Due to the phosphate head which is hydrophilic and move toward water and the fatty acid tail which hydrophobic and repels water. 2 layers of monophospholipid Make a bilayer of phospholipid form

*Evidence that cell membrane is flexible:
1)since it has the ability to engulf/can change shape
2)when a needle is poked through it does not leave a whole and it reseals
3)when 2 cells are brought close to each other they fuse.

Cell Transpsort

1) diffusion

Is the movement of substances from high concentration to low concentration down the concentration gradient.
Is a passive process does not require metabolic energy
It depends on the kinetic energy of molecules, which results in the random movement of molecules.

2) osmosis

Is the diffusion of water molecules.
Is a passive process does not require energy from ATP
Needs a partially permeable membrane.

—>> isotonic solution (Same concentration as cell)

Plants.	Animals

No movement in or out of cell

—>> hypertonic solution (low water potential)

Plants.	Animals
Water molecules diffuse out of cell By osmosis. Cells get flaccid/plasmolyzed.	Water molecules diffuse out of cell By osmosis Cells shrink

—>> hypotonic solution (high water potential)

Plants.	Animals
Water molecules diffuse into cell. By osmosis. Cells get turgid.	Water molecules diffuse into cell By osmosis. Cells burst

*water potential: how many water molecules is there to bind with other molecules.

3) facilitated diffusion

Diffusion of large, ionic molecules using channel proteins
Passive process
Requires channel proteins, which have hydrophilic R-groups between them allowing ions to move through.
Channel proteins are specific and might work in pairs.

4) active transport

Movement of molecules from high concentration to low concentration
Needs energy from respiration, since carrier proteins need energy to change shape and shuttle molecules across membrane
Requires carrier proteins
Needs partially permeable membrane
Moves molecules in one direction only
(Only occurs in living organisms, cells carrying active transport have many mitochondria to provide ATP)

5) endocytosis

When large material needs to get in cell
Ex: white blood cell engulfing bacteria (phagocytosis)
Needs energy supplied by ATP
A vesicle binds to cell membrane and intakes material then enters back to cell

6)exocytosis

When large material need to get out of cell
Ex:releasing of digestive enzymes
Needs energy supplied by ATP
A vesicle fuse with cell membrane and releases material

Gas exchange

A) small organisms:

Have large surface area to volume ratio

As a result needed oxygen is directly absorbed from surroundings through their outer body surface

B) large organisms:

Have small surface area to volume ratio

As a result needed oxygen needs to travel long distances to reach all cells which is not fast nor efficient, so they developed exchange systems

*have high metabolic rate to control their body temperatures at active.

Factors affecting gas exchange

1) Surface area: positive correlation; more particles are available to be exchanged at equal time

2) concentration: positive correlation: particles will move faster to maintain the concentration gradient

3) thickness: negative correlation, longer distance for particles move so slower diffusion

** Fick's law of diffusion:

$$\text{rate of diffusion} = \frac{\text{surface area} \times \text{concentration difference}}{\text{Thickness of exchange membrane}}$$

Properties of efficient gas exchange system:

1- large surface area
2- thin membrane
3- rich blood supply, to maintain a steep concentration gradient
4- moist surface ,some gases are in a solution
5- permeable surface

Human gas exchange system:

-Is located in the chest

-Connected to outside surroundings by nasal cavity;which is covered in tiny hairs and mucus to prevent entrance of pathogens. Also haas moist surfaces and rich capillary supply to ensure air is entering the lung at correct temperature and does not change internal environment of lungs

-Alveoli is the main site of gas exchange which is made of a single layer of epithelial cells and is folded to increase surface area and is surrounded by a rich supply of blood capillaries.

-Blood has high concentrations of carbon dioxide and air has high concentration of oxygen this creates a steep concentration gradient for diffusion to occur quickly

Gas	Percentage of gas in:		
	Inspired air	Alveoli air	Expired air
Oxygen	20.7	13.2	14.5
Carbon dioxide	0.04	5	3.9
Nitrogen	78	75.6	75.4
Water Vapour	1.24	6.2	6.2

Ventilation:

Breathing out: Passive process

- Diaphragm relaxes and become dome-shaped
- External intercostal muscles relax
- Ribs move down and in
- Volume of chest cavity decrease and pressure increases and air is forced out

[coughing is forced exhalation where energy is used to pull chest cavity further down to increase pressure in it]

Breathing in: active process

- Diaphragm contracts and flatten
- External intercostal muscles contract
- Ribs move upward and outwards
- Volume of chest cavity increases and pressure decreases and air is forced in

Enzymes

What is an enzyme?
Globular Protien that acts as a biological catalyst without being involved or used up in reaction
By lowering activation energy of a reaction

Types of enzymes:
1- anabolic —> build up
2- catabolic —> break down

How do enzymes work?
Substrate + enzyme ⇌ Enzyme-substrate complex ⇌ Enzyme + product
* enzymes are specific, have a particular shaped active. Site which only fits certain substrate or a certain group of substrates

Factors affecting enzyme activity
(1) temperature:
as temperature increase, rate of reaction increase, as molecules gain more kinetic energy and move faster increasing frequency of enzyme hitting substrate forming more enzyme-substrate complex also, molecules have more energy meaning number of successful collision increase, until it reaches optimum temperature then rate of reaction decrease as enzyme,which is a protein, denatures losing its quaternary and tertiary structure; this changes the active site of an enzyme so no more enzyme-substrate complex forms and enzyme loses its ability to catalyst the reaction

(2) pH:
change in pH affects the formation of hydrogen bonds and disulfide bonds which hold the 3D structure of protein together; changing shape of active site

(3) enzyme concentration:
as enzyme concentration increase rate of reaction increase as more active sites are available to form more enzyme-substrate complex form up to a point until it is no longer the limiting factor

(4) substrate concentration:
as substrate concentration increase rate of reaction increase up until all enzyme is saturated, all active sites are occupied, then an increase in substrate will not affect the rate of reaction

Write an investigation to show the effect of enzyme concentration on the rate of reaction:

$$H_2O_2 \xrightarrow{\text{Catalase}} H_2O + O_2$$

1) bring 5 test tubes, put same volume an concentration of substrate {H2O2} in each test tube
2) add same volume but different concentration of enzyme {catalase} to each test tube
3) put all 5 test tubes at the same temperature use thermostatically controlled water bath to maintain temperature
4) leave all test tubes for an equal. Period of time
5) measure the dependent variable {oxygen gas} using a gas syringe
6) calculate rate of reaction. By dividing volume of oxygen over time
7) repeat the investigation at each different concentration and get mean

Write an investigation to show the effect of temperature on the rate of reaction:

$$\text{Lipid} \longrightarrow \text{Fatty acid} + \text{glycerol}$$

1) bring 5 test tubes, put same volume an concentration of substrate {lipid} in each test tube
2) add same volume and concentration of enzyme {lipase} to each test tube
3) put each test tubes at a different temperature using a thermostatically controlled water bath to maintain temperature
4) leave all test tubes for an equal Period of time
5) measure the dependent variable {fatty acid } by measuring pH of test tube
6) calculate rate of reaction By dividing pH over time
7) repeat the investigation at each different concentration and get mean

Structure of DNA and RNA

Function of mono-nucleotides:

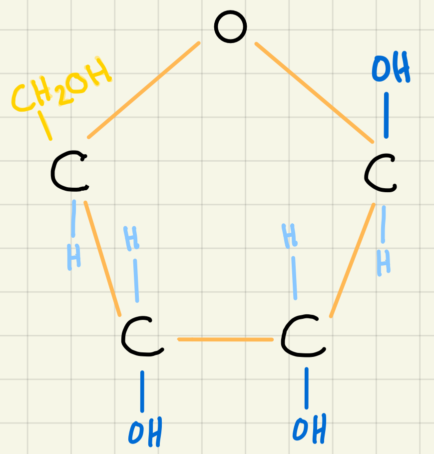
1- provide energy in form of ATP (adenosine triphosphate)

2- provide bulding block of inheritance in form of DNA (deoxyribonucleic acid) and RNA (ribonucleic acid)

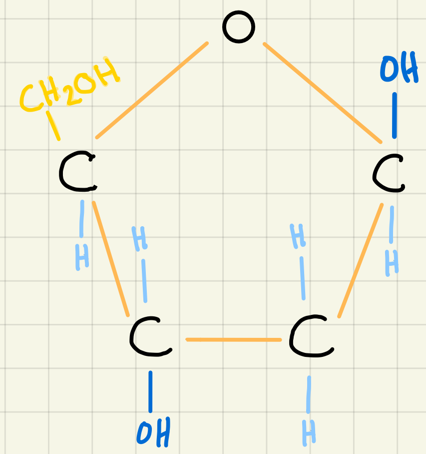
Parts of nucleotides:

1- a 5-carbon pentode sugar

[DNA sugar is deoxyribose which has one less oxygen atom than RNA ribose sugar]



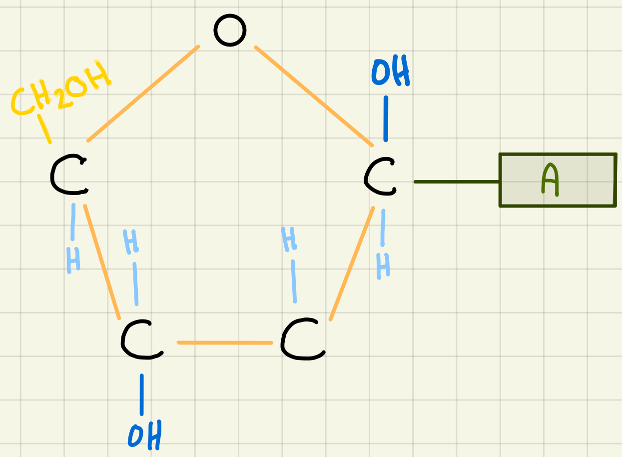
⟶ Ribose

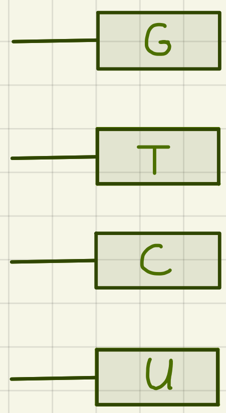


⟶ Deoxyribose.

2- a nitrogen containing base

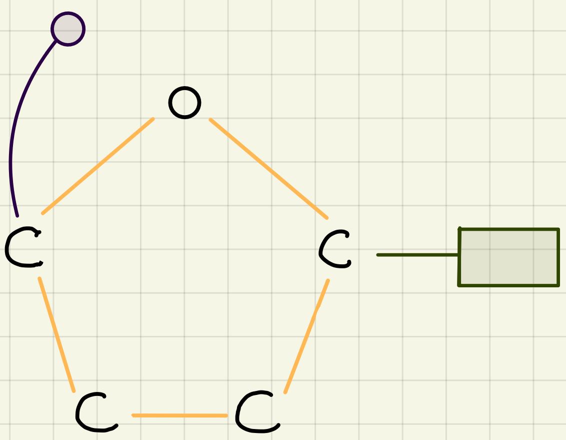
[Purine based, has 2 nitrogen containing rings. Ex: adenine and guanine
Pyrimidine base, has 1 nitrogen contains ring. Ex: cytosine, thymine and uracil]





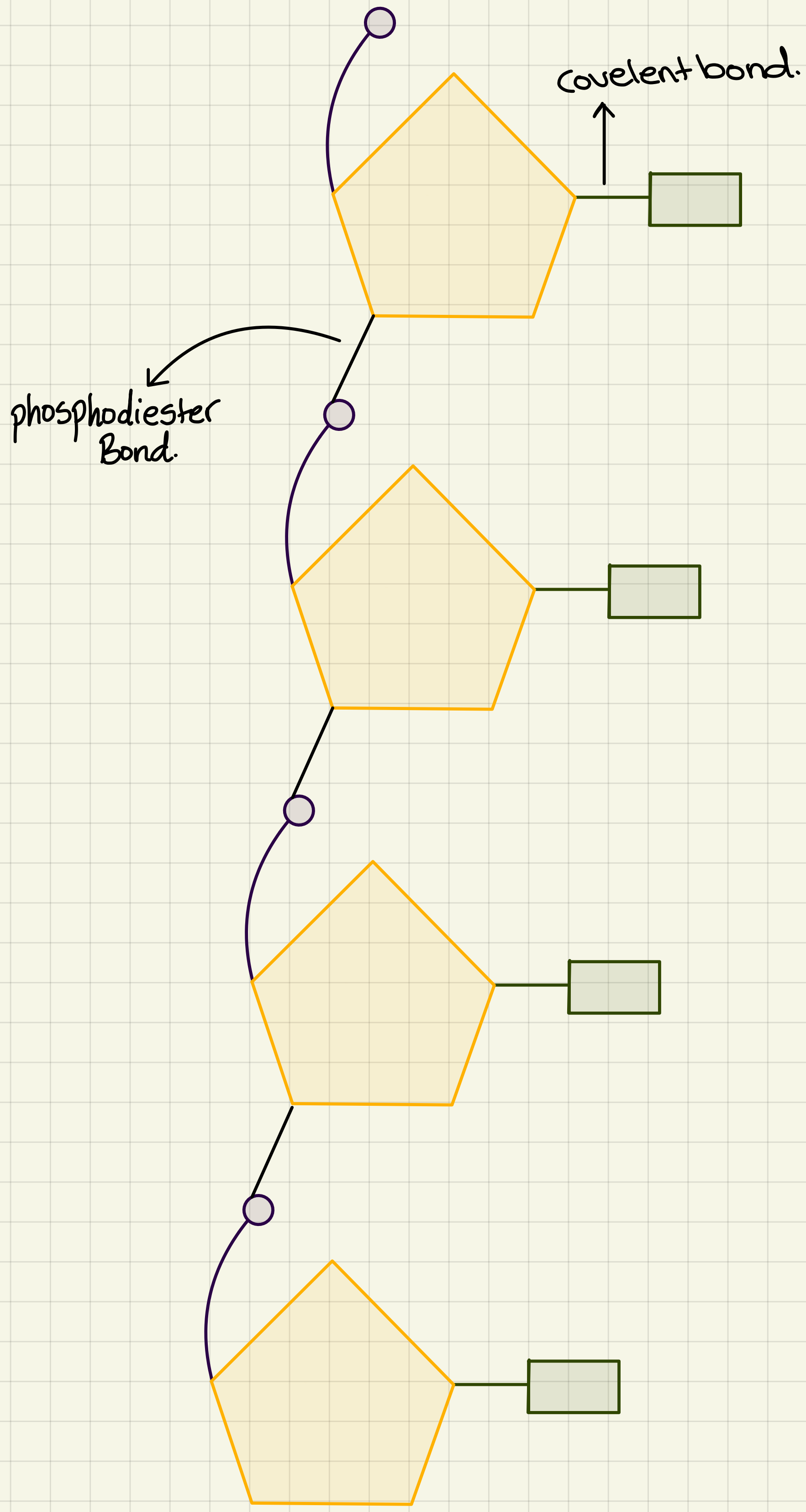
3- a phosphate group

[makes nucleotide acidic and carry a negative charge]



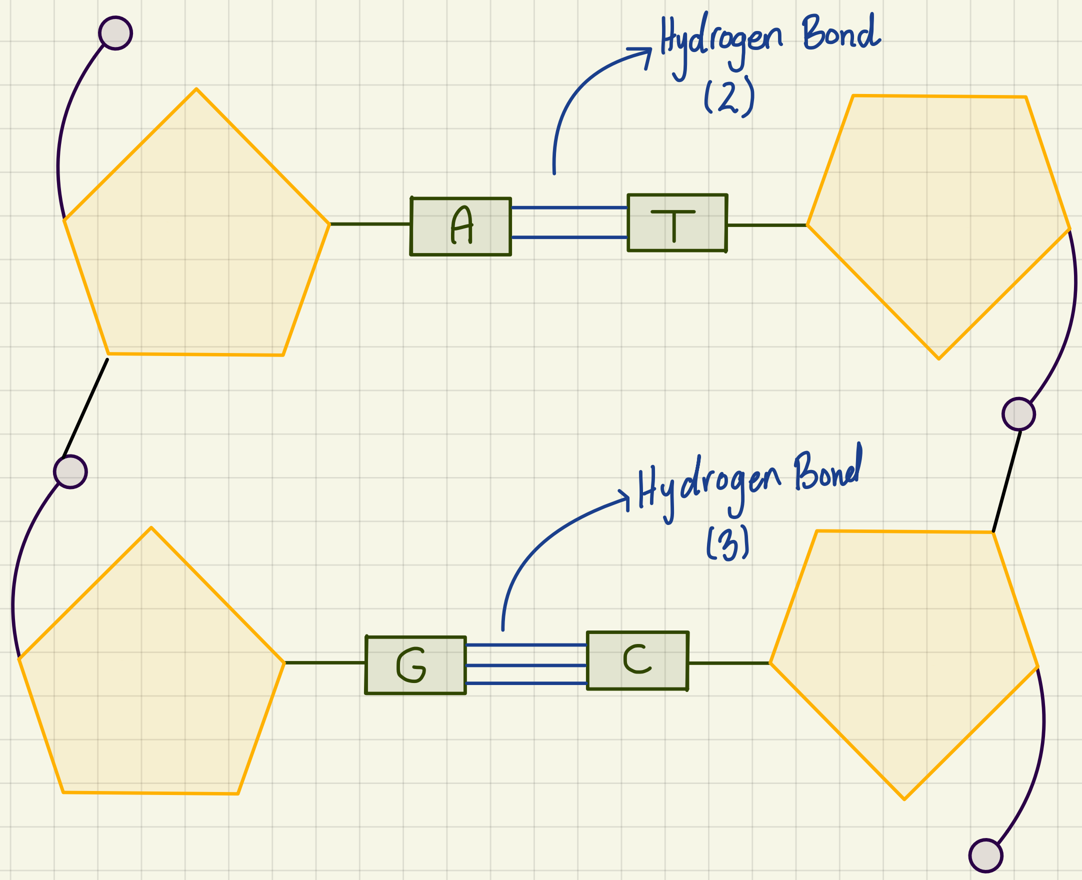
What is a polynucleotide?

A polymer of mono-nucleotide linked together by phosphodiester bond through condensation reaction, which act a store of genetic information

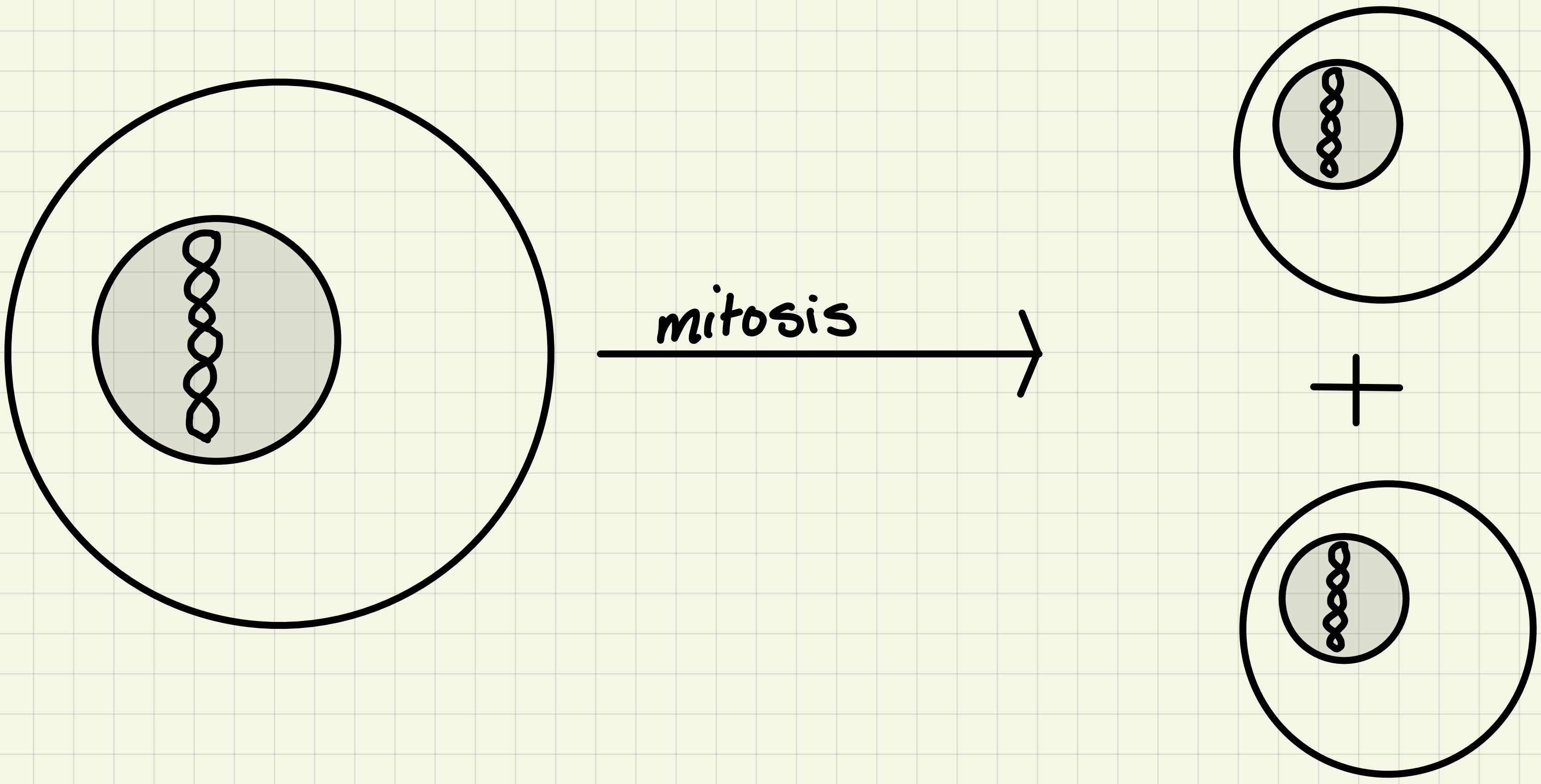


RNA molecules form of a single strand of polynucleotides and can either fold into complex shapes held by hydrogen bonds, or remain as a long strand
(Bases: A + U and G + C)

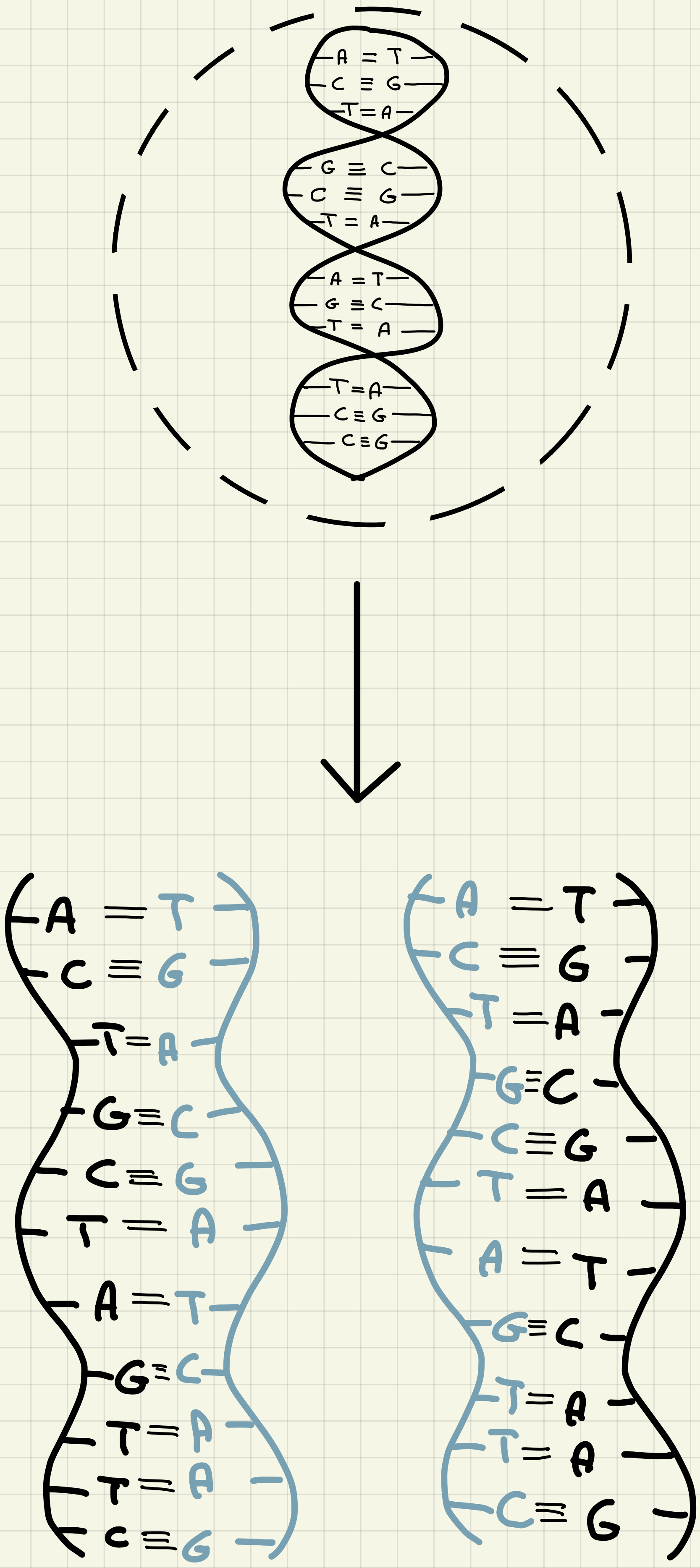
DNA is made from two polynucleotides strands twisted around each other in a complementary base pair order to form a double helix
(Bases: A + T and G + C)



*Always one small organic group with one large organic group to maintain distance so when strand bends it does not break



Before mitosis, DNA replication takes place in nucleus.



DNA replication:

-Double helix unzips into 2 strand by help of enzyme DNA helicase (polymerase)

-Free DNA nucleotides in nucleus line up against the two strands in complementary base pairing

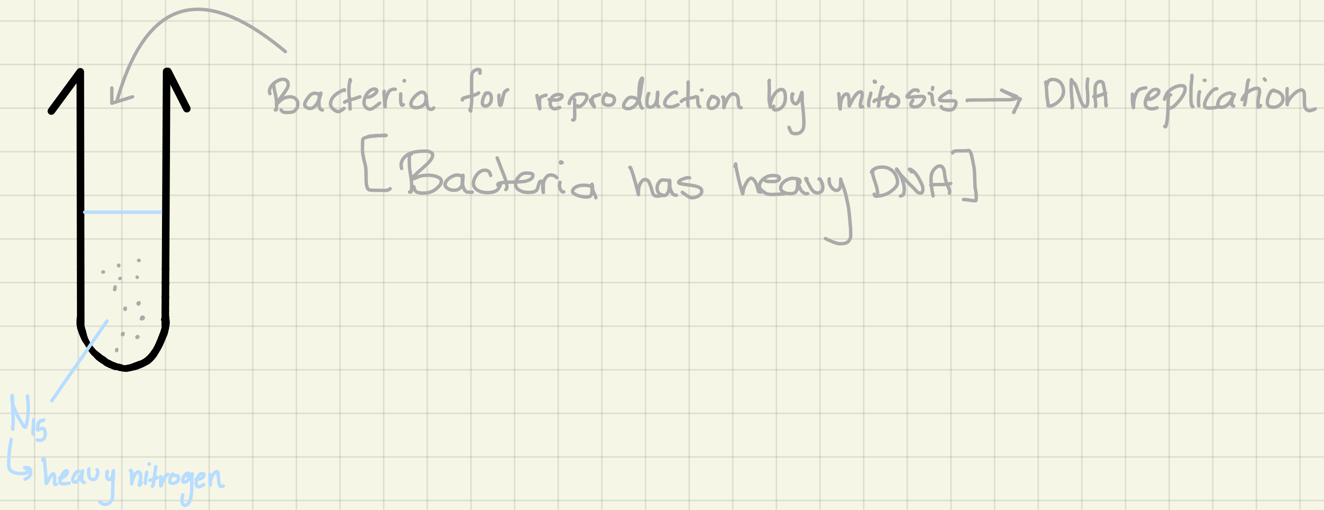
-Phophodiester bond is formed between phosphate group and sugar by condensation reaction

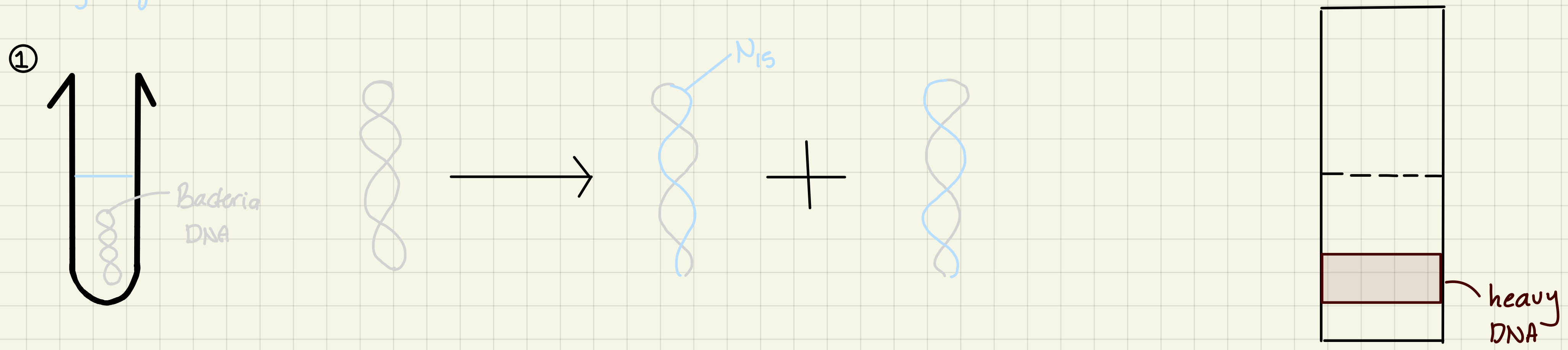
-Hydrogen bonds form between complementary nitrogen bases by help from enzyme DNA ligase

-As a result 2 new but identical to parent DNA molecules are formed

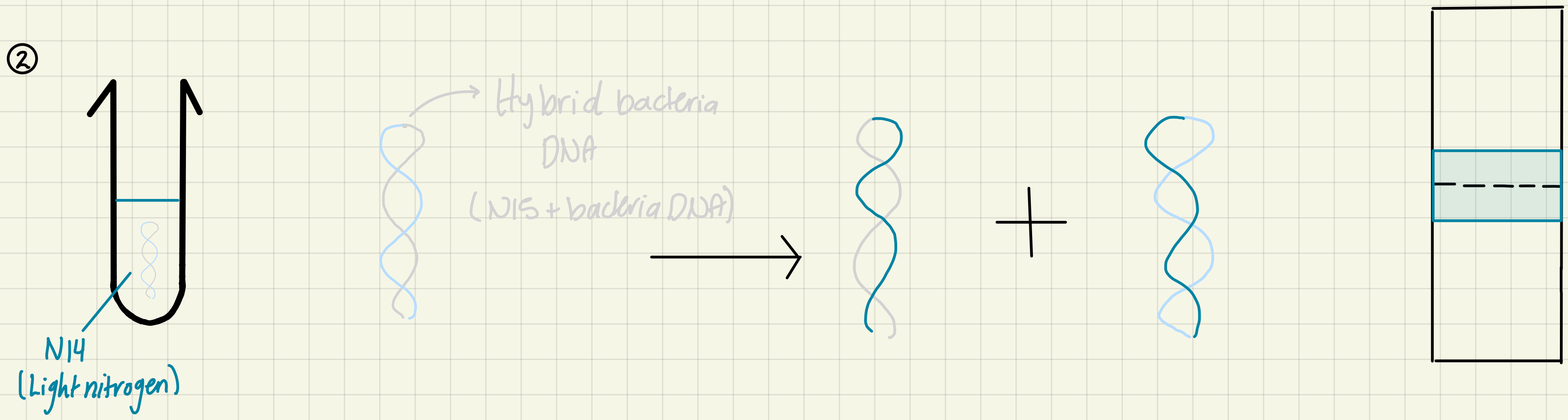
** is up to the semi-conservative theory

Evidence that DNA replication follows the semi-conservative theory
{Meselson and Stahel theory}

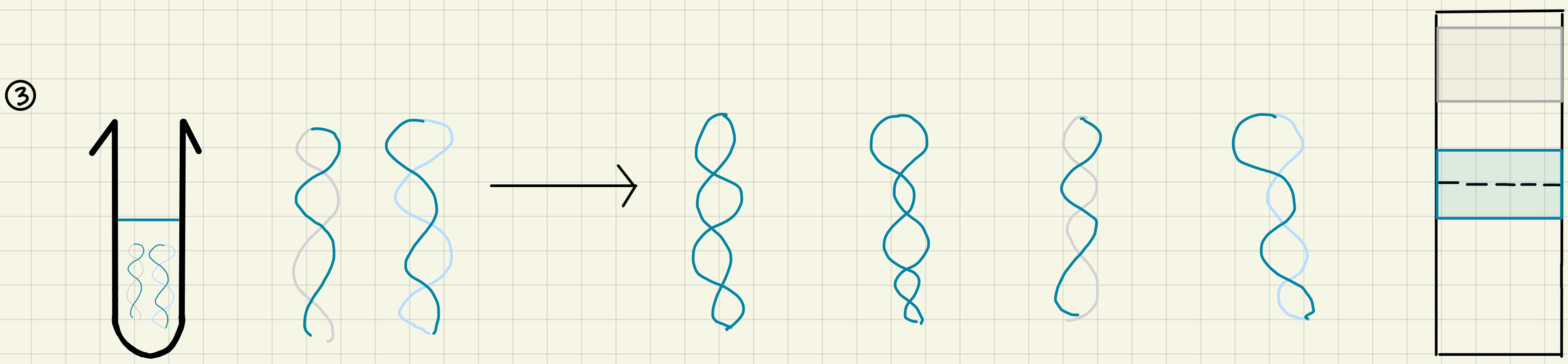




Heavy N15 + heavy DNA —>> very dense
When put in a centrifuge it sinks (one bottom band)



Both have 1 heavy + 1 light —>> med density
It settles in middle in a centrifuge



2(light +heavy) + 2(light+ heavy) —>> 1 med density + 1 light density bands

Protein synthesis

Gene: sequence of bases which codes for a polypeptide

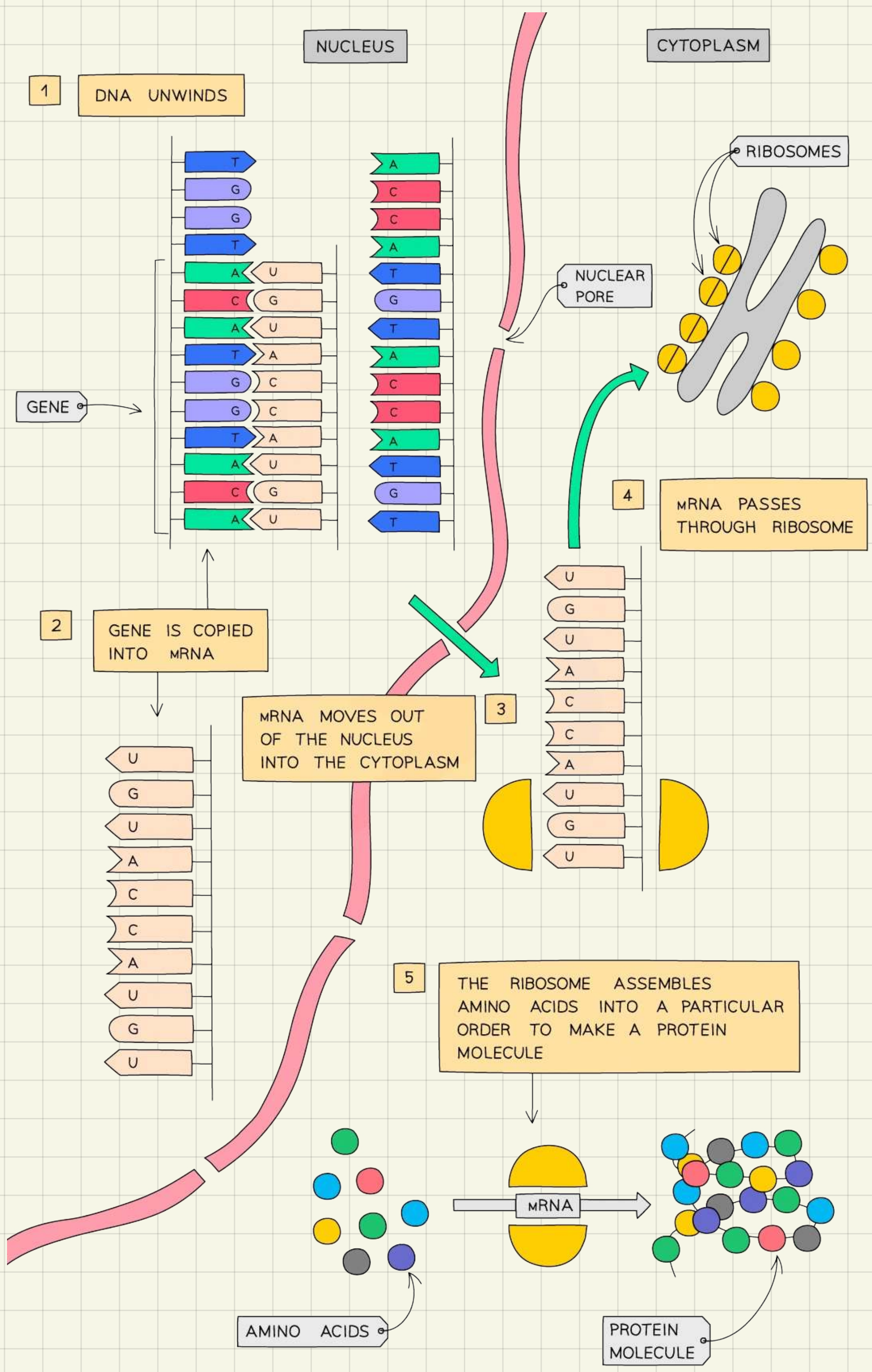
2 stages:

a) transcription in nucleus:
-DNA molecule unzips and DNA strands separate and hydrogen bonds break by RNA polymerase
-RNA mono-nucleotides line up against antisense strand which acts as a template
-in a complementary base pairing
- phosphate group and sugar get joined by condensation reaction and form phosphodiester bond
-mRNA carries copy of gene and leaves nucleus to cytoplasm. And attaches to ribosome

b) translation in cytoplasm in ribosomes:
-tRNA carries specific amino acid and line up against mRNA
- in a complementary base pairing
-interaction between tRNA and mRNA
-peptide bond is formed between 2 adjacent amino acids by condensation reaction
- forming sequence of amino acid;primary structure of protein
-tRNA detach itself and carry another amino acid

* DNA = code
* mRNA = codon
*tRNA = anticodon

→ interaction between codon and anticodon to give time for amino acids to bind to each other



• Genetic code:
Each amino acid has genetic code

What is the nature (properties) of amino acid?

1) triplet code: each three bases code for 1 amino acid
[there are 20 amino acids in nature, and only 4 bases which code for amino acid. If the genetic code is a double cod, 2 bases, only 16 amino acids would exist. So the minimum number of bases needed to code for one amino acid is 3]
2) degenerate: one amino acid with multiple genetic codes
[due to mutation, one base is changes to another yet the amino acid stays the same; this minimizes the effect of mutation.]
3) non-overlapping: each base is only a part of one amino acid
[minimize effect of mutation. As if one base is mutated only one amino acid is affected]

*start codon = first amino acid which binds to mRNA; helps other amino acids to bind to each other
* stop codon = stops the process of protein synthesis, does not code for an amino acid.

Mutation: change in the DNA base sequence,
↳ changes the sequence of amino acid
↳ Changes sequence and type of R-groups
↳ Changes folding of polypeptide
↳ Changes protein produced

(1) Point Mutation:

(a) substitution:
Affects only one amino acid, which will result in:
→ Still codes for same amino acid
→ Codes for another amino acid
→ Codes for a stop codon (shortens length of polypeptide)

(b) deletion:
One base gets deleted and all bases under it shift upward; affects all amino acids under it. Negative effect increase as you go up the strand

(c) insertion:
One base gets added and all bases under it shift downward; affects all amino acids under it. Negative effect increase as you go up the strand

(2) Chromosomal mutation:
*most dangerous: whole chromosome gets mutated, added or lost. (Down syndrome)

Why does a change in DNA sequence changes the properties of an enzyme?
Changes in amino acid sequence due to change in DNA bases. So R-groups change → bonds change → folding of polypeptide change → shape of active site changes → substrate no longer fits/binds active site

Inheritance

Gene: sequence of bases which codes for a polypeptide

Allele: a variant of a gene

Locus: location of gene on chromosome

Dominant: characteristic that is expressed in an individual wether they are homozygous or heterozygous

Recessive: characteristic that is expressed in an individual only when they are homozygous

Co-dominance: when two different alleles coding for a specific characteristic are fully expressed in the phenotype of a heterozygous individual

Homozygous: where both alleles coding for a specific characters are the same

Heterozygous: where the two alleles coding for a specific characters are different

Phenotype: the physical traits of an individual [is a result of inherited genes + environmental factors]

Genotype: the genetic make up of an organism

— **Homologous pairs**: arrangement of chromosomes inherited from each parent according to genes (a particular gene inherited from mother and father are found at the same locus in each chromosome)
* all genes carried in the chromosomes of a homologous pairs are the same, might have different alleles, except for the sex chromosome

Monogenic crossing

True breeding: both parents are homozygous, both are dominant or both recessive

Ex: BB + BB → 100% BB

Monohybrid breeding: both parents are homozygous but one Is dominant homozygous and the other is recessive homozygous

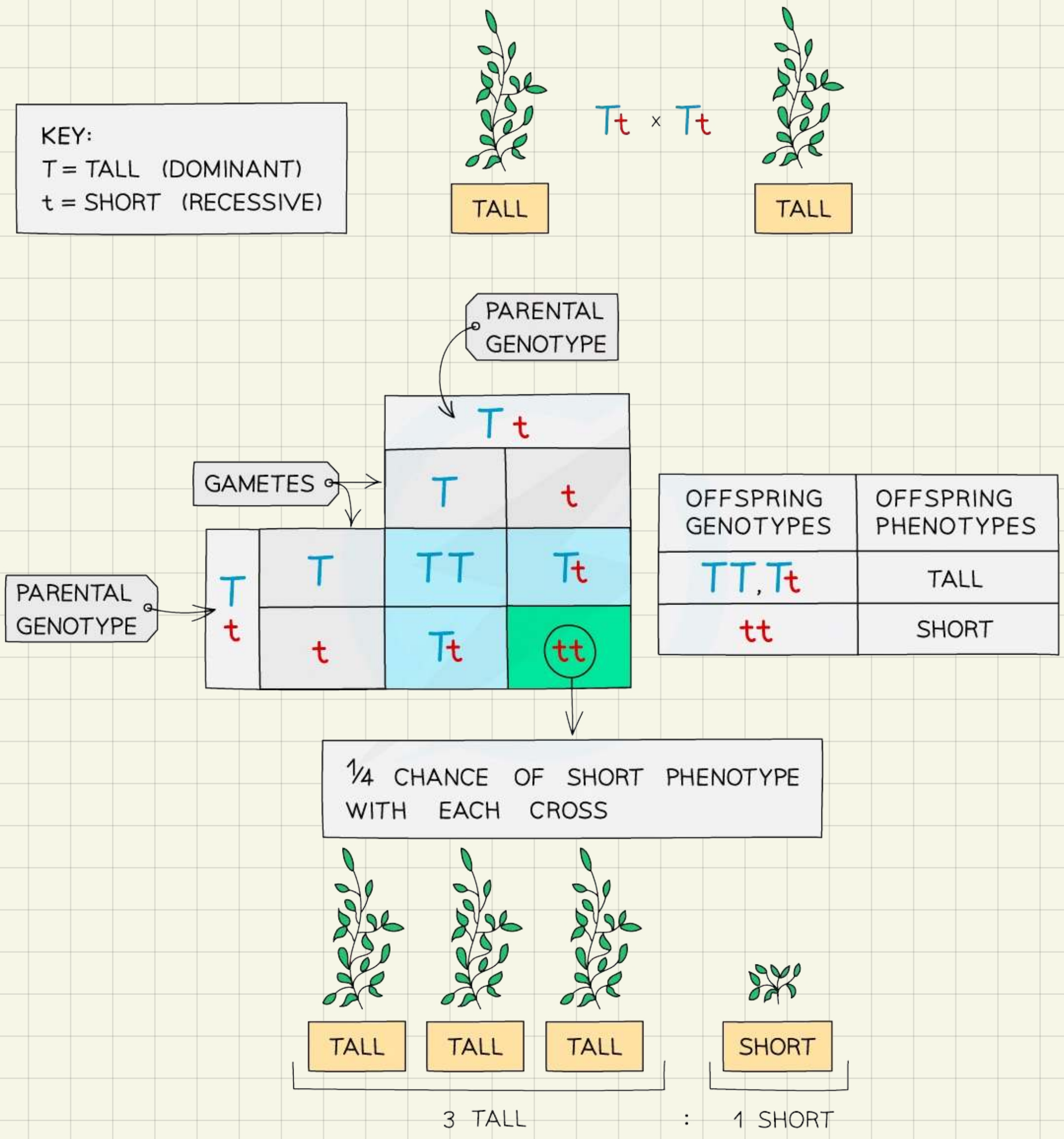
Ex: BB + bb → 100% Bb

A punnet square is used to predict the offsprings when crossing two parents

Helps in predicting pass down of inherited diseases over a long period of time

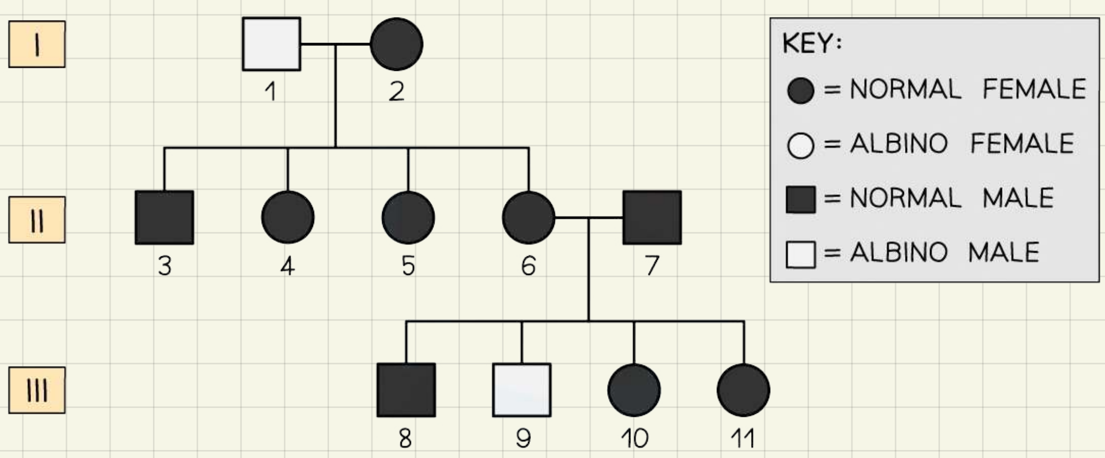
Test cross: to reveal the parental genotype

Ex: brown fur is dominant, what is the genotype of a brown rabbit? Breed with white rabbit [homozygous recessive/ bb] if all offsprings are brown then brown rabbit has genotype of BB [homozygous dominant] if half of the offsprings are white then brown rabbit has a genotype of Bb [heterozygous]



* if an offspring of two identical phenotype parents showed a different phenotype then both parents are heterozygous

In a genetic diagram include, parents' phenotype, genotype and gamete + offsprings' genotype and phenotype



Pedigree digram, helps in tracking mutation and understanding genetic history of a family

Why are fruit flies and pea plants used for genetic experiments?
- no ethical issues
- short life span, to see results of crosses/ mutation quickly
- fast reproduction
- easy and cheap to grow
- produce large number of offsprings, to produce statistically relevant results
- have clear and easily distinguishable features

[disadvantages: random mating, not all eggs are fertilized, insects can escape easily]

What are the sampling errors associated with genetic experiments?
- reproduction is a result of chance
- some offsprings might die before being sampled, miscarriages
- inefficient sampling techniques, such as escape of insects

Sex determination:

Autosomal chromosomes (22 pairs) : all chromosomes (23 pairs) except one non-homologous pair which is the sex chromosome pair (1 pair).
Sex chromosome, X chromosome and Y chromosome carry information about an individual's gender
In humans females are homogametic since both chromosomes in her eggs are X chromosome
And human males are heterogametic since they have both X and Y chromosome in their sperm
X chromosome is much larger and contains more bases, as a result codes for more protein than Y chromosome

* despite differences in Y and X chromosome they are similar enough to pair up during cell division

Sex linkage:

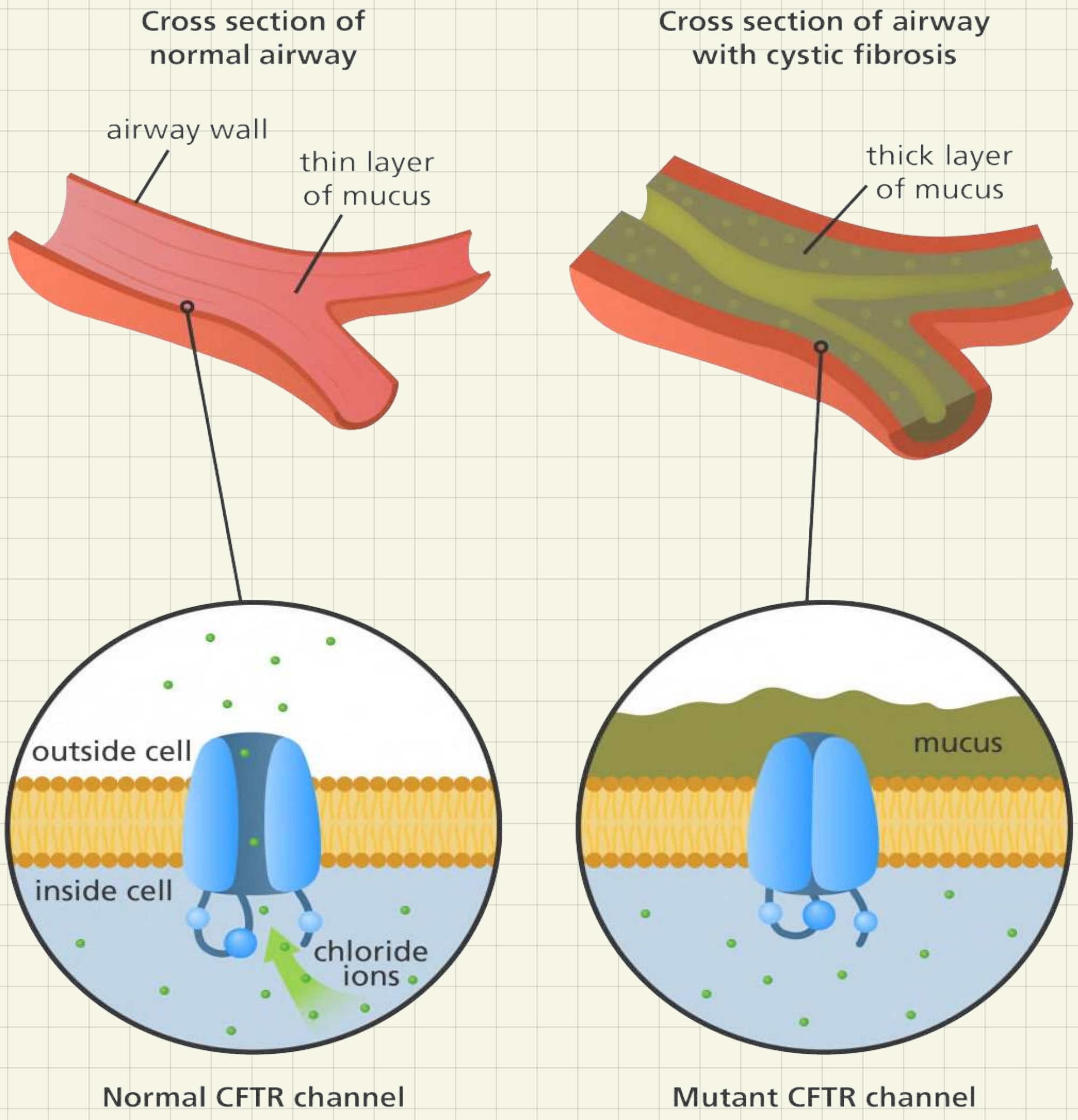
Genes carried on X chromosome are sex-linked genes
A gene passed on X chromosome from mother to son will always be expressed even if the allele is recessive since their is no corresponding allele on the Y chromosome, as Y chromosome only carries genes which code for male traits.
As a result any mutation on X chromosome will affect the son's phenotype wether in a minor or life-threatening way
→ Red-Green color blindness
Is a result of a recessive mutation that occurs in the X chromosome genes
As a result it more common in men than in women
Women can be healthy, affected or carriers while men can only be healthy or affected
[most sex-linked diseases are more common in men than in women and are passed to sons by their mothers]

Cystic fibrosis

Mutation in Cl^- (CFTR) channel proteins [recessive mutation]

↳ Result: faulty channels so, Cl^- can't move out of cell and water moves into cell leaving mucus thick and sticky

* Cl channel proteins → CFTR channel proteins



Na^+ channel proteins and Cl^- channel proteins work in pairs, if Cl^- channel proteins are open then Cl^- moves outside cell so, water potential in cell increases. As a result water moves by osmosis from cell, high water potential to surrounding, low water potential. This water mixes with mucus making it fluid. In mutated CFTR channels Cl doesn't move outside cell so, water potential in cell decreases. As a result water moves by osmosis from surrounding to cell. Leaving thick sticky mucus which can't be removed easily by cilia.

* channel proteins are made from long chain of amino acids so mutation can occur easily

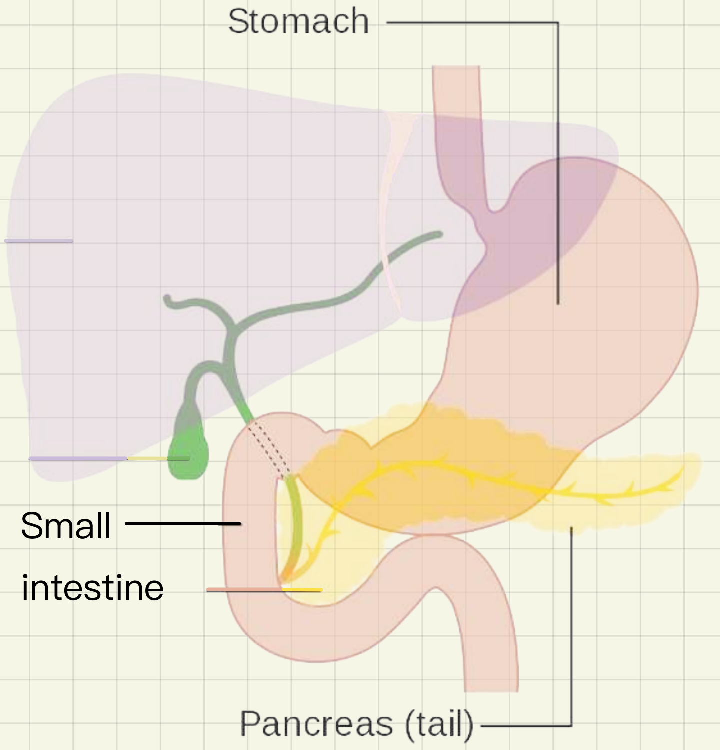
It effects:

① Respiratory system:

Mucus is thick and sticky, can't be removed by cilia or coughing [hardness in breathing], mucus narrows and blocks trachea so less air enters lungs, consequently gas exchange level decrease, since diffusion gradient is less steep and distance between air and blood capillaries increases so gases diffuse slowly, less oxygen reaches cells and less energy is produced [tiredness and fatigue] + mucus is a good environment for bacteria to grow [patients repeatedly get lung infections]

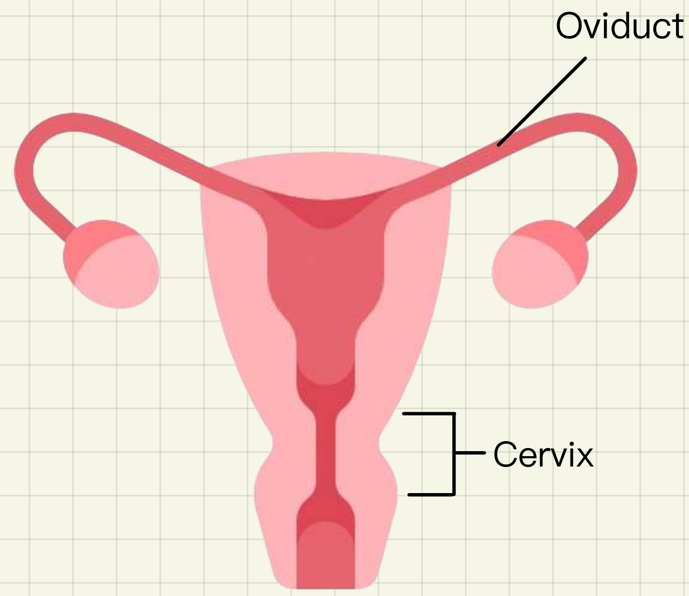
② Digestive system:

-pancreatic duct has goblet cells which have CFTR channel proteins and produce mucus.
-CFTR channels are faulty leading to having thick sticky mucus which blocks the pancreatic duct so enzymes can't move to small intestine.
-Food in small intestine is not digested and can't get absorbed [weight loss and malnutrition]
-Enzymes stuck in pancreas may digest islets of Langerhans, insulin producing cells, so no more insulin is produced [patients may suffer from diabetes]



③ Reproductive system:

-Oviduct and cervix contain goblet cells which have CFTR channels and produce mucus
-CFTR channels are faulty leading to having thick sticky mucus which blocks oviduct and cervix
- ovum can't be released into uterus since oviduct is blocked and can't be fertilized
- sperm can't enter uterus since cervix is blocked and can't fertilize ovum
[patients become infertile]
-In men with cystic fibrosis they either lack vas deferens or mucus blocks the vas deferens so no or little sperm can leave the testis.



* Explain how mutation leads to thick sticky mucus?
Change in DNA base sequence which codes for CTFR channel protein, amino acid sequence changes, R-groups type and position change so bonds change and three dimensional folding changes; making a faulty CFTR channel proteins. Cl can not move into cell; decreasing water potential in cell so, water moves into cell leaving mucus thick and sticky

Cystic fibrosis is a genetic disease that is inherited or caused by mutation with no permanent treatment

Genetic screening:

To test for inherited diseases

Tt X Tt → 25% of children will be affected (thalassemia)

Prenatal genetic screening:

Carried out on fetus cells to test for potential genetic diseases

1) amniocentesis:

Sample is taken from amniotic fluid.
At 16 week (4 months).
Fluid and fetal cells are separated in centrifuge
Then cultured for 2 weeks to get a larger sample of fetus cells.
Sample goes through DNA analysis looking for faulty genes.

Disadvantages:
- high risk of miscarriage since it's carried late in pregnancy {1%}
- result need a long time to be processed (2-3 weeks)
- can only be carried relatively late in pregnancy, making it very difficult for parents to terminate pregnancy if necessary

2) chronic villus:

Sample is taken from placenta
At 10 weeks (2 months)
*since placenta already has a large sample of fetal cells no need to culture sample
Sample goes through DNA analysis looking for faulty genes.

Disadvantages:
- Risk of miscarriage {1-2%}
- any faulty gene located on paternal X chromosome can't be detected

*Advantages over amniocentesis:
- less physical (and emotional) traumatic for mother if abortion is necessary, why? because it is carried at much earlier
- faster results

Preimplantation genetic screening:

Through IVF: egg and sperm are fertilized outside body after few cell divisions a singular cell is removed from each embryo and tested for faulty genes, healthy embryos are placed into mother uterus to implant and grow

Ethical issues:

- risk of miscarriage → killing of living organisms
- not 100% accurate :

    False positive, might lead to an abortion of a healthy baby

    False negative, might lead to having a sick baby
- who will take the decision, mother or father
- religious/ societal/ personal/ political beliefs differ from a person to another
- might discover other gene mutations by mistake
- in IVF there is an issue surrounding removing the carrier (but not affected) embryos. Parents may want to keep it but communities tilt toward removing the faulty allele from the gene pool
- traumatic to families to know they have an inherited diseases in their genes

Counseling:

To help people with assessing their situation, coming to term with it and give couple options available to them and rise awareness toward the future their heading to. They will work with parent to choose what they believe is right for their situation within their framework of morals, family, religion, social beliefs and traditions